

Derivatives Pricing Project

LLSMS2225

11-01-2026

Professor: Frédéric Vrins

Assistant : Rodolphe Vanderveken



LOUVAIN
School of Management

GROUP 8

Jean de Biolley - 55612100

Evrard de Marchant – 51102000

Gaëtan Franck - 71692200

Basile Kummeler - 90592200

Introduction	3
1. Construction and intuition of the trade.....	3
2. Valuation Methodologies and Comparative Pricing.....	7
2.1 Pricing with Monte Carlo Simulations	8
2.2 Numerical Integration Pricing	9
2.2.1 The Absolute Payoff	10
2.2.2 The Quadratic Payoff	10
2.3. Analytical Derivation Pricing.....	11
2.3.1 The Absolute Payoff	11
2.3.2 The Quadratic Payoff	13
2.4. Analysis and graphs of the sensitivity of the option price	14
2.4.1 Calculation of the sensitivity	14
2.4.2 Analytic Derivation of the Greeks.....	15
1. Delta (Δ)	15
Delta of the Absolute Payoff	15
Delta of the Quadratic Payoff	15
2. Gamma (Γ)	16
Gamma of the Absolute Payoff	16
Gamma of the Quadratic Payoff	17
3. Theta (Θ).....	17
Theta of the Absolute Payoff.....	17
Theta of the Quadratic Payoff	18
Link to the BSM PDE Relationship	18
4. Vega (ν)	19
Vega of the Absolute Payoff.....	19
Vega of the Quadratic Payoff	21
2.4.3 Graphical Analysis and Interpretation	22
2.5 The Delta (Δt) Analytical Computation.....	23
2.5.1 The Absolute Payoff Delta	23
Behavior of the Absolute Delta	24
2.5.1 The Quadratic Payoff Delta.....	25
Behavior of the Quadratic Delta.....	25
2.6. Analysis of the Payoff, Option Price, Profit and Delta functions	26
2.6.1 Payoff Functions	26
2.6.1 Option Price Functions	27
2.6.2 Profit Functions.....	28
2.6.3 Delta Functions.....	29
3. Dynamic Replication of Option Payoffs	31
3.1 Implementation of a Dynamic Replication Strategy	31
3.1.1 Simulation of the underlying asset paths.....	31
3.1.2 Replication principle: delta hedging.....	31
3.1.3 Analytical pricing and delta formulas.....	32
Absolute payoff	32
Quadratic payoff.....	32
3.1.4 Portfolio decomposition	33
3.1.5 Replicating Portfolio Dynamics.....	33
Initialization	34
Update between two rebalancing dates	34
Termination at maturity	34
3.1.6 Replicating portfolio dynamics over multiple rebalancing periods	35

3.2 Impact of Transaction Costs on Replication	35
3.2.1 Effect on Portfolio Dynamics and Hedging Error.....	35
3.2.2 Rebalancing Frequency and Cost–Accuracy Trade-off	37
3.3 Comparison of Replication Performance	37
3.3.1 Interpretation of the replication error statistics.....	38
General effects of rebalancing frequency and transaction costs	38
Mean replication error ($\text{Mean}(\varepsilon)$).....	38
Standard deviation of the replication error ($\text{SD}(\varepsilon)$).....	39
MAE - Mean Absolute Error	40
RMSE - Root Mean Squared Error.....	40
3.3.2 Interpretation of replication error quantiles	41
Lower quantiles: downside risk (1% and 5%)	41
Central tendency: median (50%)	41
Upper quantiles: favorable outcomes (95% and 99%)	42
3.3.3 Graphical analysis	42
Density plots of the replication error ($n = 252$)	42
Boxplots of the replication error.....	43
RMSE vs number of rebalancing intervals.....	44
Mean transaction cost vs number of rebalancing intervals	45
3.4 Chapter 3 Conclusion	46
4. Final Assessment and Strategic Recommendation	47
Final Product Recommendation.....	47
Bibliography.....	49

Introduction

Financial markets are historically dominated by directional bets, where investors seek to profit from the specific rise or fall of an asset's price. In this traditional view, volatility is seen merely as a risk factor. Moving beyond traditional directional strategies, investors now actively treat volatility as a tradable asset class in its own right. The concept of trading volatility was initially proposed by Brenner and Galai (1989) to provide better tools for measuring market risk and creating effective hedging instruments. Building on their work, the Chicago Board Options Exchange (CBOE) introduced the VIX index in 1993 to serve as a benchmark for expected short-term volatility. This eventually transformed volatility into a widely accessible asset class.

This report addresses a specific mandate from a client desiring a deal on an asset (S) for which she anticipates heavy variations over the coming years. Indifferent to the direction of the movement, she seeks to capitalize solely on the magnitude of the deviation. To fulfill the request, we analyze and compare two distinct payoff structures; an absolute payoff and a quadratic payoff. The absolute payoff is a linear approach that generates returns proportional to the move. The quadratic payoff is a convex approach that generates returns proportional to the square of the move.

Our analysis proceeds in three stages. First, we examine the construction and the intuition of these trades, demonstrating that while the absolute payoff corresponds to a simple combination of vanilla options, the quadratic payoff requires a more complex structure. We utilize the Carr-Madan (1998) static replication strategy to show how a portfolio of OTM options can synthesize a convexity exposure that no single option can provide. Second, we perform a rigorous pricing analysis under the Black-Scholes framework, cross-validating results via Monte Carlo simulations, numerical integration, and analytical formulas. Finally, we evaluate the operational and hedging risks from the perspective of the issuer. By implementing a dynamic Delta-hedging strategy, we assess the impact of discrete-time rebalancing and proportional transaction costs on the final profit and loss, ultimately determining the practical feasibility of both strategies in a realistic market environment.

1. Construction and intuition of the trade

To satisfy the client's mandate, capitalizing on magnitude while ignoring the direction, we use two possible payoff structures. These payoff function must be symmetric around a central strike price K and we treat the asset price at maturity S_T purely as deviation from it. The two payoff structures are respectively the absolute payoff and the quadratic payoff:

$$\psi_T^a = |S_T - K| \tag{1}$$

$$\psi_T^q = (S_T - K)^2 \tag{2}$$

The absolute payoff generates a positive, linear return for every step the asset moves away from the strike. This strategy is called a long straddle and involves purchasing one European call and one European put option with the same strike price K and maturity T . By summing the payoffs of the two options we obtain the absolute payoff structure :

$$\text{European call : } \psi_{call} = \max(0, S_T - K)$$

$$\text{European put : } \psi_{put} = \max(0, K - S_T)$$

$$\psi_T^{total} = \max(0, S_T - K) + \max(0, K - S_T)$$

If $S_T \geq K$, the Call option is In-the-Money and pays $S_T - K$, while the Put option pays 0. On the other side if $S_T < K$, the Call option pays 0, but the Put option pays $K - S_T$. In both cases the payoff is given by :

$$\psi_T^{total} = |S_T - K|$$

This confirms that the deal can be constructed with two vanilla options. The straddle provides a constant slope of 1 so when the stock moves \$10 up, the call pays \$10. If it moves \$10 down, the put pays 10\$. This linearity is the baseline against which we will compare the more aggressive quadratic strategy.

The quadratic payoff represents a convex parabola and because vanilla options possess linear payoff functions, it is geometrically impossible to perfectly replicate a smooth parabolic curve using a combination of a finite number of calls and puts. However, the payoff can be replicated with a strategy known as ‘Static Replication’. As demonstrated in the paper “Towards a Theory of Volatility Trading” by Carr and Madan (1998), any twice-differentiable payoff function can be covered by a portfolio of Out-of-the-Money options across a continuum of strike prices.

The Carr-Madan formula relies on the Taylor expansion with an integral remainder. We start with the Fundamental Theorem of Calculus that tells us we can reconstruct the value of a function $f(S_T)$ by starting at a reference point, in this case our strike price $f(K)$ and adding up all the incremental changes (the slope) that happened between K and S_T :

$$f(S_T) = f(K) + \int_K^{S_T} f'(t)dt \quad (3)$$

This equation tells us that the final price is just the starting price plus the accumulation of all slopes along the way. However, this form relies on the first derivative, which corresponds to directional risk. In finance, we know that options prices are driven by convexity (curvature), not just direction. To capture this curvature, we need to introduce the second derivative into the equation by integrating the integral term $(\int_K^{S_T} f'(t)dt)$ of equation (1) by parts:

$$u = f'(t) \Rightarrow du = f''(t)dt$$

$$dv = dt \Rightarrow v = t + C = -(S_T - t)$$

$$\int u \, dv = uv - \int v \, du$$

$$\int_K^{S_T} f'(t) dt = [-f'(t)(S_T - t)]_K^{S_T} - \int_K^{S_T} -(S_T - t)f''(t) dt \quad (4)$$

The mathematical trick is to choose the constant of integration specifically as $C = -S_T$ so that $v = t + C = -(S_T - t)$. This choice is mathematically valid and necessary for two reasons. First, it forces the boundary *term* $[uv]$ to equal zero at the upper limit $t = S_T$, significantly simplifying the expansion. Next, it introduces the term $(S_T - t)$, which matches the payoff structure of a call option, allowing us to interpret the integral as a portfolio of options. Evaluating the boundary terms and rearranging the equation, we arrive at the Carr-Madan Spanning Formula:

$$f(S_T) = \underbrace{f(K)}_{\text{Bond}} + \underbrace{f'(K)(S_T - K)}_{\text{Delta Hedge}} + \underbrace{\int_0^K f''(k)(K - S_T)^+ dk + \int_K^\infty f''(k)(S_T - K)^+ dk}_{\text{Portfolio of Options}} \quad (5)$$

This derivation allows us to interpret the Taylor expansion through a purely financial lens, effectively decomposing any payoff function into a static portfolio of three distinct instruments. The first term, $f(K)$, represents a risk-free bond, corresponding to the fixed cash value held if the price remains exactly at the strike. The second term, $f'(K)(S_T - K)$, acts as a forward contract or a static Delta hedge, capturing the linear exposure to the underlying asset's direction. Finally, the integral remainder term, which mathematically accounts for the curvature of the function, corresponds exactly to a portfolio of Out-of-the-Money Call and Put options. Crucially, the second derivative $f''(k)$ serves as a weighting function, dictating the precise quantity of options required at each strike to replicate the desired convexity.

When applying this general formula to our specific quadratic contract, the equation simplifies a lot. When we evaluate the expansion components at the strike price K , the bond term effectively disappears, as the function's value at the strike is zero ($f(K) = (K - K)^2 = 0$). Similarly, the delta term vanishes because the first derivative, $f'(S_T) = 2(S_T - K)$, evaluated at the strike ($S_T = K$) is zero. However, the convexity term determined by the second derivative, remains strictly positive with a constant value of $f''(k) = 2$. Substituting these values into the spanning formula eliminates the linear components entirely, leaving only the option portfolio:

$$f(S_T) = \int_0^K f''(k)(K - S_T)^+ dk + \int_K^\infty f''(k)(S_T - K)^+ dk \quad (6)$$

$$\psi_T^q = \int_0^K 2 \cdot P(k) dk + \int_K^\infty 2 \cdot C(k) dk \quad (7)$$

While the quadratic deal cannot be constructed with a single pair of options, it can be perfectly replicated by holding a static portfolio containing every available OTM Put and Call option, each weighted by a constant factor of 2. In a real-world, discrete market where strikes are spaced by ΔK rather than a continuum, we approximate this interval by a sum. We would hold a weight of $2 \times \Delta K$ contracts for each strike to synthesize the exposure.

To rigorously compare the absolute and the quadratic strategies, we visualize their payoffs at maturity T as a function of the underlying asset price S_T . Figure 1 illustrates the structural difference between the two. The absolute payoff function exhibits a v-shape, characteristic for a straddle. As the asset price moves away from the strike price, the returns grow linearly. The slope is constant; equal to -1 for $(S_T < K)$ and to $+1$ for $(S_T > K)$. The quadratic payoff function

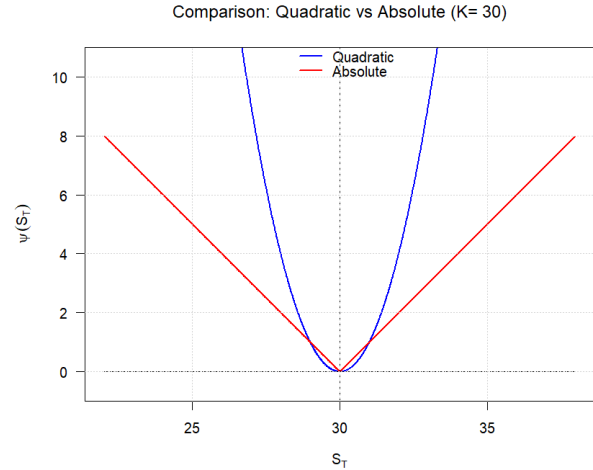


Figure 1: Payoff Comparison

exhibits a u-shape parabola. The returns grow exponentially as the asset price moves away from the strike price. Unlike the linear model, the slope is not constant. It steepens as the distance from the strike increases, reflecting the positive convexity of the position.

The crossover points of the two payoff function are at $|S_T - K| = 1$, they determine the exact point where the two contracts will have the exact same payoff. When the asset price remain close to the strike ($|S_T - K| < 1$), the absolute payoff is strictly superior to the quadratic payoff. On the other side, when there is more significant volatility and the asset price deviates a lot from the strike price ($|S_T - K| > 1$) the Quadratic Payoff outperforms the linear strategy.

Consequently, the relative value of the two contracts (V_0^a versus V_0^q) acts as a direct view on expected volatility over the life of the trade. In a low volatility environment, the market anticipates the asset price to remain within a narrow range around the strike. If the probability density function is concentrated in the region where deviations are less than 1 unit ($|S_T - K| < 1$), the straddle will theoretically command a higher price than the quadratic contract because the probability of realizing the explosive squared returns is negligible. Conversely, in a high volatility environment, there is a significant probability of large movements where deviations far exceed 1 unit. The value of the quadratic contract is dominated by the potential for explosive payoffs. The convexity premium kicks in, making the quadratic deal significantly more expensive.

Choosing the best contract basically boils down to a simple question: “Are we betting on a move smaller than \$1 or bigger than \$1?”. Under standard market parameters where the asset price is $S_0 = 30$, a typical standard deviation move (e.g., 10% or 20%) implies a price

change of \$3 to \$6. Since this is significantly larger than the critical crossover of 1 unit, we intuitively expect the Quadratic Price (V_0^q) to be higher than the Absolute Price (V_0^a) under normal market conditions. The market naturally prices in the squared upside of the expected move.

2. Valuation Methodologies and Comparative Pricing

Following the theoretical construction of our two products, we now transition to determining their fair market value. To ensure the robustness of our results, we use 3 approaches to cross-validate our results: Monte Carlo simulation, numerical integration and analytical derivation.

Consistent with the Black-Scholes-Merton framework, all pricing is conducted under the risk-neutral measure (P^*), assuming a frictionless market where the underlying asset follows a Geometric Brownian Motion (GBM). Within this idealized environment, we assume that the risk-free rate ($r = 4$ and volatility ($\sigma = 15$) remain constant, we postulate that the asset pays no dividends and that trading is continuous with no arbitrage opportunities or transaction costs during the initial pricing phase. By establishing these parameters for our initial price $S_0 = 30$ and maturity $T = 1$, we provide a stable foundation to evaluate the contract prices.

2.1 Pricing with Monte Carlo Simulations

To determine the fair value of both the absolute (1) and quadratic (2) contracts, we first employ Monte Carlo simulations. This method approximates the risk-neutral expectation of the payoff by simulating a large number of possible future asset price trajectories. Consistent with the Black-Scholes framework and the project assumptions, we model the underlying asset price S_t as a Geometric Brownian Motion (GBM). Under the risk-neutral measure P^* , the stochastic differential equation (SDE) governing the asset is:

$$dS_t = rS_t dt + \sigma S_t dW_t^*, \quad (8)$$

where W_t^* is a Standard Brownian motion under P^* .

To simulate the price at maturity T , we solve this SDE using Itô's Lemma. The exact solution allows us to step directly from $t = 0$ to $t = T$ without discretization error for the final value:

$$S_T = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma W_T^* \right) \quad (9)$$

Since $W_T^* \sim \mathcal{N}(0, T)$, we can express the random shock using a standard normal variable $Z \sim \mathcal{N}(0, 1)$:

$$S_T^{(i)} = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z^{(i)} \right) \quad (10)$$

According to the fundamental theorem of asset pricing, in a market free of arbitrage opportunities, the price of any derivative V_0 is equal to the expected value of its future payoff, discounted at the risk-free rate, provided the expectation is taken under our specific risk-neutral measure (P^*). $\Psi(S_T)$ represents the payoff function at maturity T and r is the constant risk-free rate:

$$V_0 = e^{-rT} E^*[\Psi(S_T)] \quad (11)$$

It is important to clarify that we do not use the real-world probability measure (P). In the real world, risky assets earn a risk premium ($\mu > r$) to compensate investors for uncertainty. However, because derivatives can be perfectly replicated using the

underlying asset, their prices should not depend on the investors' risk preferences or the asset's physical drift μ . Therefore, we price the derivative in a theoretical risk-neutral world where all assets, regardless of their risk, are expected to grow at the risk-free rate r .

By the Law of Large Numbers (LLN), we can approximate this continuous expectation by averaging the payoffs from a finite number (N) of independent simulations. As $N \rightarrow \infty$, the standard error of the estimator decays to zero and the sample mean converges to the true theoretical price:

$$V_0 \approx \widehat{V}_{MC} = e^{-rT} \frac{1}{N} \sum_{i=1}^N \Psi(S_T^{(i)}) \quad (12)$$

To approximate the price, we execute a Monte Carlo algorithm using $N = 1,000,000$ independent sample paths. The process involves generating standard normal random variables $Z^{(i)} \sim \mathcal{N}(0,1)$ to compute the terminal asset. The resulting prices are $V_0^{(a)} = 3.64$ and $V_0^{(q)} = 22.7415$.

The quadratic contract commands a significantly higher premium (≈ 22.75) compared to the absolute contract (≈ 3.64). This aligns with our intuition from chapter 1 that the quadratic function assigns an exponentially higher weight to large deviations. While the probability of extreme moves is low, their contribution to the expected payoff is massive due to the squaring of the distance $(S_T - K)^2$, whereas the absolute payoff $|S_T - K|$ only scales linearly.

2.2 Numerical Integration Pricing

While Monte Carlo simulation provides a robust stochastic estimate, especially with $N=1,000,000$, we can achieve a more stable and precise fair value by evaluating the risk-neutral expectation deterministically. This approach involves solving the pricing integral directly against the probability density function of the terminal asset price.

To compute the expectation $E^*[\Psi(S_T)]$ we must integrate the payoff over the probability density of S_T . However, integrating directly over a log-normal distribution is hard and can be numerically unstable due to the complex shape of its density function. To simplify this, we perform a change of variable introduced in section 2.1. We know that the Brownian motion term W_T^* follows a Normal distribution $\mathcal{N}(0,T)$. We can standardize this by defining a standard normal variable $u \sim \mathcal{N}(0,1)$, such that $W_T^* = \sqrt{T} \cdot u$. This allows us to express the final stock price as a direct function of the standard normal outcome u :

$$S_T(u) = S_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}u\right) \quad (13)$$

Now when computing the expectation, instead of a complex log-normal integral, we solve a simpler integral against the standard Gaussian density function $\phi(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$.

2.2.1 The Absolute Payoff

We first value the contract with the absolute payoff (1). Mathematically, the price is the integral of all possible payoffs weighted by their probability density function. We can compute the expected payoff by integrating the profit function against the standard normal probability density function $\phi(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$. This transformation allows us to visualize the pricing problem as calculating the area under a curve defined by the product of the payoff profile and the Gaussian probability:

$$V_0^a = e^{-rT} \int_{-\infty}^{+\infty} \underbrace{|S_T(u) - K|}_{\text{Payoff for outcome } u} \cdot \underbrace{\phi(u)}_{\text{Probability of } u} du \quad (14)$$

We can solve this integral using adaptive quadrature. Unlike Monte Carlo simulation, which approximates the value by averaging random samples, numerical integration calculates the precise area under the probability curve by discretizing the domain into a fine grid and summing the areas. This method eliminates the randomness of the simulation and obtain a stable and highly accurate result for the expected value. Using the market parameters, the numerical integration yields a price of:

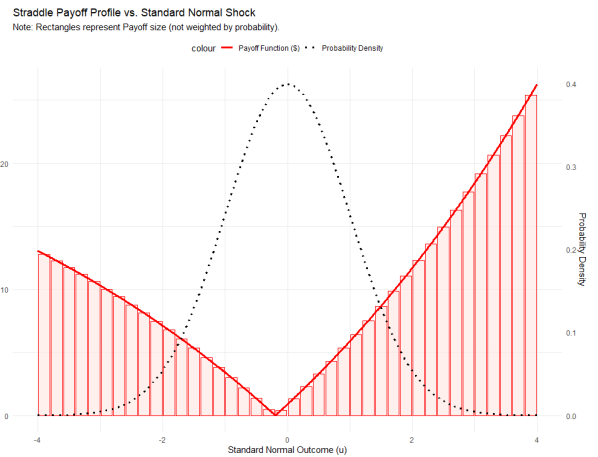


Figure 2: Absolute Payoff Numerical Integration

$$V_0^a \approx 3.6408$$

2.2.2 The Quadratic Payoff

If we maintain the exact same integration framework, changing only the payoff function inside the integral we obtain this:

$$V_0^q = e^{-rT} \int_{-\infty}^{+\infty} \underbrace{(S_T(u) - K)^2}_{\text{Quadratic Payoff}} \cdot \underbrace{\phi(u)}_{\text{Probability Density}} du \quad (15)$$

This integral formulation reveals why the quadratic contract is so expensive. For standard normal realizations u near zero, when the asset price remains close to the strike K , the squared payoff term $(S_T(u) - K)^2$ is small. Even though the probability $\phi(u)$ is high, the total contribution to the integral is small. Conversely, for extreme positive or negative values of u , representing tail events, the squared payoff expands exponentially. Although the probability density $\phi(u)$ associated with these extreme outcomes is low, massive magnitude of the squared payoff overpowers it. The area under the curve in these tail regions is surprisingly large, effectively inflating the total price. This mathematically

confirms our intuition that the value of the quadratic contract is dominated by the potential for extreme moves.

This behavior highlights a critical advantage of using numerical integration over stochastic simulation for this specific contract. A standard Monte Carlo simulation might rarely sample these extreme tail events, unless the number of iterations is massive, potentially underestimating the true value if the random samples fail to hit the far tails. Numerical integration, by contrast, deterministically evaluates the product of the payoff and probability density across the entire infinite domain, ensuring that the massive value hidden in these low-probability but high-impact zones is fully captured. By summing the probability-weighted squared payoffs across the entire distribution, we obtain the following price:

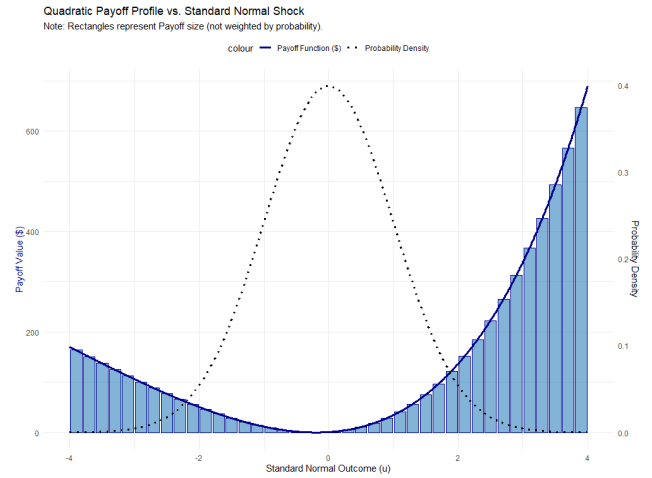


Figure 3: Quadratic Payoff Numerical Integration

$$V_0^q \approx 22.7555$$

This confirms that the massive premium over the linear payoff is driven by the convexity of the contract structure.

2.3. Analytical Derivation Pricing

While the previous sections provided stochastic and numerical estimates, we now we derive analytical solutions for both contracts. These formulas rely on the risk-neutral moments of the Geometric Brownian Motion (GBM).

2.3.1 The Absolute Payoff

We start the analytical computation by inserting the decomposed absolute payoff, into the sum of a European Call and a European Put with the same strike K and maturity T , into our discounted expected value of the payoff at maturity under the risk-neutral measure (11):

$$|S_T - K| = (S_T - K)^+ + (K - S_T)^+ \quad (16)$$

$$V_0 = e^{-rT} E^{P^*} [|S_T - K|] \quad (11)$$

$$V_0^a = e^{-rT} E^{P^*} [(S_T - K)^+] + e^{-rT} E^{P^*} [(K - S_T)^+] = C_0 + P_0 \quad (17)$$

By the linearity of the expectation operator, the price of the absolute payoff is simply the sum of the vanilla option prices.

Under the Black-Scholes-Merton assumptions, where the underlying follows a Geometric Brownian Motion, the analytical prices for C_0 and P_0 are given by:

$$C_0 = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2) \quad (18)$$

$$P_0 = Ke^{-rT} \Phi(-d_2) - S_0 \Phi(-d_1) \quad (19)$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function and the terms d_1, d_2 are defined as:

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \quad (20)$$

$$d_2 = d_1 - \sigma\sqrt{T} = \frac{\ln(S_0/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \quad (21)$$

Substituting these components, the final analytical equation of the absolute payoff price is given by:

$$V_0^a = [S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2)] + [Ke^{-rT} \Phi(-d_2) - S_0 \Phi(-d_1)] \quad (22)$$

To compute the price, we begin by calculating the d -statistics. Given that the contract is at-the-money ($S_0 = K$), the logarithmic term $\ln(S_0/K)$ vanishes, simplifying the calculations:

$$d_1 = \frac{(r + \frac{1}{2}\sigma^2)T}{\sigma} = \frac{0.04 + \frac{1}{2}(0.15^2)}{0.15} = \frac{0.04 + 0.01125}{0.15} = \frac{0.05125}{0.15} \approx 0.3417$$

$$d_2 = d_1 - \sigma\sqrt{T} = 0.3417 - 0.15 \approx 0.1917$$

Using standard normal cumulative distribution tables, we determine: $\Phi(d_1) \approx \Phi(0.3417) \approx 0.6337$ and $\Phi(d_2) \approx \Phi(0.1917) \approx 0.5760$. Following the symmetry of the Gaussian distribution, the values for the out-of-the-money components are found to be $\Phi(-d_1) \approx 0.3663$ and $\Phi(-d_2) \approx 0.4240$.

These values are then integrated into the Black-Scholes pricing formula alongside the annual discount factor $e^{-rT} \approx 0.9608$, to obtain the call premium (C_0) and the put premium (P_0):

$$C_0 = 30 \cdot 0.6337 - 30 \cdot 0.960789 \cdot 0.5760 \approx 19.011 - 16.600 \approx 2.41.$$

$$P_0 = 30 \cdot 0.960789 \cdot 0.4240 - 30 \cdot 0.3663 \approx 12.219 - 10.989 \approx 1.23.$$

Summing these components yields a total time-0 price for the absolute payoff of $V_0^a \approx 3.64$. This premium represents the initial cost the client must incur to establish a delta-neutral position designed to profit from significant volatility in the underlying asset.

2.3.2 The Quadratic Payoff

The pricing of the quadratic payoff (2) follows a similar logic of decomposition through the expansion of the quadratic form:

$$(S_T - K)^2 = S_T^2 - 2KS_T + K^2 \quad (23)$$

Applying the linearity of the expectation operator again, we can decompose the total option price into three distinct components:

$$V_0^q = e^{-rT} (E^*[S_T^2] - 2KE^*[S_T] + K^2) \quad (24)$$

This modular approach reduces the valuation problem to determining the first and second non-central moments of the terminal stock price under the risk-neutral measure P^* . Under the Black-Scholes framework, the terminal price S_T is defined as:

$$S_T = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma W_T \right) \quad (13)$$

The first moment, $E^*[S_T]$, corresponds to the forward price of the asset, $S_0 e^{rT}$, which accounts for the risk-neutral drift. To derive the second moment, $E^*[S_T^2]$, we square the terminal price expression. By substituting the standard normal variable $u \sim \mathcal{N}(0,1)$, where $W_T = \sqrt{T}u$, the squared terminal price is expressed as:

$$S_T^2 = S_0^2 \exp \left((2r - \sigma^2)T + 2\sigma\sqrt{T}u \right) \quad (25)$$

Given that $u \sim \mathcal{N}(0,1)$, we utilize the moment-generating property of the standard normal distribution, $E[e^{ku}] = e^{\frac{1}{2}k^2}$. By setting the constant $k = 2\sigma\sqrt{T}$, the expectation of the stochastic component becomes $\exp \left(\frac{1}{2} (2\sigma\sqrt{T})^2 \right) = e^{2\sigma^2 T}$. Substituting this result back into the expression for S_T^2 (25) yields:

$$E^*[S_T^2] = S_0^2 e^{(2r - \sigma^2)T} \cdot e^{2\sigma^2 T} = S_0^2 e^{(2r + \sigma^2)T} \quad (26)$$

By substituting the derived moments back into the pricing equation (24), we arrive at the final analytical formula for the quadratic option:

$$\begin{aligned} V_0^q &= e^{-rT} (S_0^2 e^{(2r + \sigma^2)T} - 2KS_0 e^{rT} + K^2) \\ V_0^q &= S_0^2 e^{(r + \sigma^2)T} - 2KS_0 + K^2 e^{-rT} \end{aligned} \quad (27)$$

This equation (27) highlights the price drivers: the first term represents the convexity adjustment driven by volatility, the second is a linear adjustment for the strike, and the third is the discounted strike price.

Applying the project parameters and computing the exponential terms on this final formula (27):

$$e^{(r+\sigma^2)T} = e^{0.04+0.15^2} = e^{0.0625} \approx 1.064494$$

$$S_0^2 e^{(r+\sigma^2)T} = 900 \times 1.064494 \approx 958.045$$

$$-2KS_0 = -2 \times 30 \times 30 = -1800$$

$$K^2 e^{-rT} = 900 \times 0.960789 \approx 864.710$$

Summing these components, we find the analytical price:

$$V_0 \approx 958.045 - 1800 + 864.710 \approx 22.755$$

2.4. Analysis and graphs of the sensitivity of the option price

Having established the fair value of the contracts, we now turn to risk management. To effectively manage these positions, we must understand how the option prices fluctuate in response to changes in market conditions. These sensitivities, quantified by the partial derivatives of the pricing function $V(t, S_t)$, are collectively known as the ‘Greeks’. In this section, we derive and visualize the four primary risk measures (Delta (Δ), Gamma (Γ), Theta (Θ), and Vega (v)) to assess the stability of both the absolute and quadratic payoff structures under varying market parameters (S_0, T, K, σ).

2.4.1 Calculation of the sensitivity

To facilitate the derivation of these sensitivities, we first recall the closed-form analytical pricing equations established in section 2.3. These formulas serve as the base functions $V(t, S_t)$ that we will differentiate with respect to the market parameters.

The price V_0^a is the discounted risk-neutral expectation of the straddle payoff. As derived previously, this is the sum of a European Call and Put:

$$V_0^a = C_0 + P_0 = [S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2)] + [K e^{-rT} \Phi(-d_2) - S_0 \Phi(-d_1)] \quad (22)$$

Where the standard normal probability terms are defined as:

$$d_1 = \frac{\ln(S_0/K) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

The price V_0^q is governed by the risk-neutral moments of the terminal asset price. As established earlier (27), the value is given by:

$$V_0^q = S_0^2 e^{(r+\sigma^2)T} - 2KS_0 + K^2 e^{-rT} \quad (27)$$

2.4.2 Analytic Derivation of the Greeks

We now calculate the four primary risk measures by computing the partial derivatives of the pricing functions $V(t, S_t)$.

1. Delta (Δ)

Delta measures the rate of change in the option price when the underlying asset price moves. It is defined as the first partial derivative of the pricing function with respect to the spot price. Beyond a simple sensitivity measure, we can understand Delta as a representation of the moneyness of the option. For instance, a Delta of 0.5 implies that the option price will increase by 50 cents for every \$1 increase in the underlying asset. In the context of standard options, this value effectively serves as a proxy for probability that the option will expire In-The-Money (ITM). Consequently, an ATM option typically carries a Delta near 0.5, while values higher than 0.5 indicate the option is ITM, and lower values indicate it is Out-Of-The-Money (OTM). For put options, the interpretation remains the same but with the opposite sign (e.g., an ATM put has a Delta of -0.5).

Delta of the Absolute Payoff

For the absolute payoff, which is constructed as a sum of a long Call and a long Put, the Delta is derived by summing the Deltas of the individual components. Using the Put-Call parity relationship ($V_0 = C_0 + P_0$), we compute the initial Delta by differentiating the price formula by S_0 :

$$\Delta_{call} = \frac{\partial C_0}{\partial S_0} = \Phi(d_1) \quad (28)$$

$$\Delta_{put} = \frac{\partial P_0}{\partial S_0} = -\Phi(-d_1) = -(1 - \Phi(d_1)) \quad (29)$$

Summing equation (28) and equation (29), we derive the final expression:

$$\Delta^a = \Delta_{call} + \Delta_{put} = \Phi(d_1) + [\Phi(d_1) - 1] = 2\Phi(d_1) - 1 \quad (30)$$

Since the standard normal cumulative distribution function $\Phi(d_1)$ ranges strictly from 0 to 1, the Delta of the absolute payoff is mathematically bound between -1 and 1. At the inception of the trade ($t = 0$), if the option is At-The-Money, we obtain a Delta of approximately 0. This confirms that we are initially Delta-neutral: small movements in the underlying price do not impact the total value of the position. However, this neutrality is temporary. If the asset price rises, the Call becomes more valuable than the Put, pushing the Delta into positive territory. Conversely, if the price falls, the Put dominates, shifting the Delta to negative.

Delta of the Quadratic Payoff

The risk profile of the quadratic payoff is fundamentally more aggressive. By differentiating the pricing formula with respect to the asset price S_t , where $\tau = T - t$ represents the time remaining to maturity, we obtain:

$$V_t^q = S_t^2 e^{(r+\sigma^2)\tau} - 2KS_t + K^2 e^{-r\tau} \quad (31)$$

$$\Delta^q = \frac{\partial V_t^q}{\partial S_t} = 2S_t e^{(r+\sigma^2)\tau} - 2K \quad (32)$$

Unlike the straddle, this Delta is not bound by probabilities; it can easily exceed 1 because the payoff grows much faster than the underlying price itself. A crucial observation from this formula is the linear impact of the parameters: all other things being equal, a change in the spot price S_t or the strike K has a linear impact on the Delta. This unbounded growth implies that as the market moves in the client's favor, the number of shares required to hedge the position increases linearly without a theoretical cap.

2. Gamma (Γ)

Gamma measures the rate of change of Delta with respect to the underlying asset price, representing the convexity of the option's value. Mathematically, it is defined as the second partial derivative of the value function with respect to the stock price, or equivalently, the first derivative of Delta:

$$\Gamma = \frac{\partial^2 V}{\partial S_t^2} = \frac{\partial \Delta}{\partial S_t} \quad (33)$$

In practical terms, Gamma serves as a metric for the stability of the Delta hedge. A high Gamma implies that Delta changes rapidly with even small price movements, necessitating frequent rebalancing of the replicating portfolio to remain delta-neutral. Conversely, a low Gamma indicates a stable hedge that requires less intervention.

Gamma of the Absolute Payoff

For the absolute payoff, we derive Gamma by differentiating the Delta formula established in the previous section, $\Delta^a = 2\Phi(d_1) - 1$. Applying the chain rule to the standard normal cumulative distribution function, we proceed as follows:

$$\Gamma^a = \frac{\partial}{\partial S_t} (2\Phi(d_1) - 1) = 2\phi(d_1) \frac{\partial d_1}{\partial S_t} \quad (34)$$

Simplifying the partial derivative $\frac{\partial d_1}{\partial S_t} = \frac{1}{S_t \sigma \sqrt{\tau}}$, we obtain the general analytical expression:

$$\Gamma^a = \frac{2\phi(d_1)}{S_t \sigma \sqrt{\tau}} \quad (35)$$

This formula reveals that Gamma is dependent on the current asset price S_t . As illustrated in the Gamma sensitivity graph, it exhibits a characteristic bell curve shape: it is maximized when the option is At-The-Money ($S_t \approx K$), where the uncertainty regarding the terminal payoff is highest, and decays toward zero as the asset price moves far away from the strike. This concentration of convexity implies that the risk of hedging error is most acute when the market is trading near the strike price K .

Gamma of the Quadratic Payoff

In contrast, the Gamma of the quadratic payoff demonstrates a fundamentally different risk profile. By differentiating the linear Delta function $\Delta^q = 2S_t e^{(r+\sigma^2)\tau} - 2K$ with respect to S_t , we find that the variable S_t disappears entirely from the expression:

$$\Gamma^q = \frac{\partial}{\partial S_t} (2S_t e^{(r+\sigma^2)\tau} - 2K) \quad (36)$$

$$\Gamma^q = 2e^{(r+\sigma^2)\tau} \quad (37)$$

This result indicates that the convexity of the quadratic contract is uniform regardless of the stock price. Unlike the straddle, where hedging intensity spikes near the strike, the quadratic payoff maintains a constant curvature across the entire price domain. This confirms that the hedging difficulty associated with the quadratic contract is structural and persistent, driven solely by time and volatility, rather than localized to a specific price range.

3. Theta (Θ)

Theta measures the sensitivity of the option price to the passage of time, often referred to as time decay. Mathematically, it is defined as the partial derivative of the value function with respect to time:

$$\Theta = \frac{\partial V}{\partial t} \quad (38)$$

Intuitively, as time passes, the likelihood of the underlying asset making a significant move decreases simply because there is less time to move. Consequently, the extrinsic value of the option erodes. This decay is not linear; it accelerates as the option approaches maturity, particularly for At-The-Money options where the uncertainty is highest. While Theta is typically negative for long positions, as we will observe with the straddle, this is not a universal rule as deep-in-the-money puts can exhibit positive Theta due to the interest rate effect on the strike price.

Theta of the Absolute Payoff

For the absolute payoff, we derive the Theta by summing the time sensitivities of the underlying European Call and Put options. Recall from the Black-Scholes framework that the Theta of a call and the Theta of a put, where $\tau = T - t$, are respectively:

$$\Theta_{call} = -\frac{S_t \sigma \phi(d_1)}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2) \quad (39)$$

$$\Theta_{put} = -\frac{S_t \sigma \phi(d_1)}{2\sqrt{\tau}} + rKe^{-r\tau}\Phi(-d_2) \quad (40)$$

Summing these components yields the total Theta for the straddle:

$$\Theta^a = -\frac{S_t \sigma \Phi(d_1)}{\sqrt{\tau}} - rK e^{-r\tau} (2\Phi(d_2) - 1) \quad (41)$$

This formula reveals the interaction between two competing forces. The first term $\left(-\frac{S_t \sigma \Phi(d_1)}{\sqrt{\tau}}\right)$ represents the volatility decay. For a straddle, this decay is doubled compared to a single option because both the call and put legs lose extrinsic value as time evaporates. The second term $(-rK e^{-r\tau} (2\Phi(d_2) - 1))$ represents the cost of carry and interest rate sensitivity. The behavior of this second term creates a unique dynamic when the option moves deep In-The-Money. If the asset price rises significantly ($S_t \gg K$), the Call option dominates the position. As $S_t \rightarrow \infty$, both d_1 and d_2 approach infinity, causing $\Phi(d_2)$ to approach 1. Consequently, the term $(2\Phi(d_2) - 1)$ approaches 1, and the volatility term vanishes. The Theta converges to $\Theta^a \approx -rK e^{-r\tau}$. In this scenario, the passage of time is detrimental because the holder is delaying the payment of the strike price to exercise the call, effectively losing the interest that could have been earned on that capital. Conversely, if the asset price falls significantly ($S_t \ll K$), the Put option dominates. As $S_t \rightarrow 0$, d_1 and d_2 approach negative infinity, causing $\Phi(d_2)$ to approach 0. The term $(2\Phi(d_2) - 1)$ becomes -1 , resulting in a Theta of $\Theta^a \approx rK e^{-r\tau}$. In this unique case, Theta becomes positive. Because the maximum payoff of a put is capped at K , holding the option allows the investor to delay purchasing the stock which is nearly worthless, while the present value of the strike price to be received increases as time passes.

Theta of the Quadratic Payoff

We derive the Theta for the quadratic payoff by differentiating the pricing formula (31) with respect to time. Since $\tau = T - t$, applying the chain rule gives $\frac{\partial}{\partial t} = -\frac{\partial}{\partial \tau}$. Differentiating the expression yields:

$$\Theta^q = \frac{\partial V^q}{\partial t} = -[(r + \sigma^2)S_t^2 e^{(r+\sigma^2)\tau} - rK^2 e^{-r\tau}] \quad (42)$$

Rearranging the terms, we obtain the final analytical formula:

$$\Theta^q = rK^2 e^{-r\tau} - (r + \sigma^2)S_t^2 e^{(r+\sigma^2)\tau} \quad (43)$$

The Theta for the quadratic payoff is heavily dominated by the second term, $-(r + \sigma^2)S_t^2 e^{(r+\sigma^2)\tau}$. This confirms that the position suffers from significant and accelerating time decay, particularly because the decay scales with the square of the asset price (S_t^2) and the square of volatility (σ^2). The cost of holding this highly convex position is paid in the rapid erosion of value as maturity approaches.

Link to the BSM PDE Relationship

The Black-Scholes-Merton Partial Differential Equation (PDE) links these Greeks together. The fundamental condition that a hedged portfolio earning the risk-free rate r must be arbitrage-free forces the Greeks to satisfy the following equality:

$$\frac{\partial V}{\partial t} + rS_t \frac{\partial V}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} = rV \quad (44)$$

In terms of the Greeks we just derived, this relationship is expressed as:

$$\Theta + rS_t\Delta + \frac{1}{2}\sigma^2 S_t^2 \Gamma = rV \quad (45)$$

This equation provides a powerful economic interpretation of the option's value: the time decay (Θ) plus the expected return from the stock ($rS_t\Delta$) and the convexity adjustment ($\frac{1}{2}\sigma^2 S_t^2 \Gamma$) must equal the risk-free return on the option's value (rV).

We can use this relationship to verify the consistency of our derived analytical formulas. By substituting our specific results for the quadratic contract of Δ^q (32), Γ^q (37), and Θ^q (43) into the LHS of the BSM relationship (45), we obtain:

$$\begin{aligned} & \underbrace{rK^2 e^{-r\tau} - (r + \sigma^2)S_t^2 e^{(r+\sigma^2)\tau}}_{\Theta^q} + \underbrace{rS_t(2S_t e^{(r+\sigma^2)\tau} - 2K)}_{rS_t\Delta^q} + \underbrace{\frac{1}{2}\sigma^2 S_t^2 (2e^{(r+\sigma^2)\tau})}_{\frac{1}{2}\sigma^2 S_t^2 \Gamma^q} \\ &= rK^2 e^{-r\tau} - rS_t^2 e^{(r+\sigma^2)\tau} - \sigma^2 S_t^2 e^{(r+\sigma^2)\tau} + 2rS_t^2 e^{(r+\sigma^2)\tau} - 2rS_t K + \sigma^2 S_t^2 e^{(r+\sigma^2)\tau} \\ &= r(S_t^2 e^{(r+\sigma^2)\tau} - 2KS_t + K^2 e^{-r\tau}) = rV^q \end{aligned}$$

The result matches rV^q exactly, confirming that our derivations for Δ^q , Γ^q , and Θ^q are mathematically consistent with the pricing formula.

4. Vega (v)

Vega measures the sensitivity of the option price to changes in the volatility σ . Mathematically, it is defined as the partial derivative of the value function with respect to volatility:

$$v = \frac{\partial V}{\partial \sigma} \quad (46)$$

In the standard Black-Scholes-Merton model, the stock price follows a Geometric Brownian Motion where volatility σ is assumed to be constant. However, option prices are highly sensitive to this parameter. Intuitively, an increase in volatility expands the range of possible future asset prices, increasing the probability of extreme outcomes. Since options have asymmetric payoffs (limited downside, unlimited or high upside), this wider distribution invariably increases the value of the option. When σ (or σ^2 in the quadratic strategy) increases, the price rises because the underlying asset S_t tends to deviate further from the strike K . As we will demonstrate, the quadratic contract represents a significantly more aggressive bet on volatility than the standard straddle.

Vega of the Absolute Payoff

To derive the Vega for the straddle, we first establish a useful relationship between the Vega of a Call and the Vega of a Put using Put-Call Parity:

$$C_0 - P_0 = S_0 - Ke^{-rT} \quad (47)$$

Differentiating both sides with respect to volatility σ :

$$\frac{\partial}{\partial \sigma} (C_0 - P_0) = \frac{\partial}{\partial \sigma} (S_0 - Ke^{-rT})$$

$$v_{call} - v_{put} = 0$$

Since the underlying price S_0 and the present value of the strike Ke^{-rT} are independent of volatility, the right-hand side is zero. Consequently, the Vega of the straddle is simply twice the Vega of a single Call option:

$$v_{straddle} = 2v_{call} \quad (48)$$

We calculate the partial derivative of the Call price (18) with respect to σ . Using the Chain Rule where $\frac{\partial \Phi(x)}{\partial \sigma} = \phi(x) \frac{\partial x}{\partial \sigma}$:

$$\frac{\partial C_0}{\partial \sigma} = S_0 \phi(d_1) \frac{\partial d_1}{\partial \sigma} - Ke^{-rT} \phi(d_2) \frac{\partial d_2}{\partial \sigma}$$

Recall the relationship $d_2 = d_1 - \sigma\sqrt{T}$. Differentiating with respect to σ gives:

$$\frac{\partial d_2}{\partial \sigma} = \frac{\partial d_1}{\partial \sigma} - \sqrt{T}$$

Substituting this back into the Vega equation:

$$\begin{aligned} v_{call} &= S_0 \phi(d_1) \frac{\partial d_1}{\partial \sigma} - Ke^{-rT} \phi(d_2) \left(\frac{\partial d_1}{\partial \sigma} - \sqrt{T} \right) \\ v_{call} &= \frac{\partial d_1}{\partial \sigma} [S_0 \phi(d_1) - Ke^{-rT} \phi(d_2)] + Ke^{-rT} \phi(d_2) \sqrt{T} \end{aligned} \quad (49)$$

To simplify the bracketed expression in the first term of equation (49), we rely on a fundamental identity of the BSM derivation: $S_0 \phi(d_1) = Ke^{-rT} \phi(d_2)$. We prove this by calculating the ratio of the standard normal densities $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$:

$$\frac{\phi(d_1)}{\phi(d_2)} = \frac{e^{-d_1^2/2}}{e^{-d_2^2/2}} = \exp\left(\frac{1}{2}(d_2^2 - d_1^2)\right)$$

$$d_1^2 - d_2^2 = (d_2 + \sigma\sqrt{T})^2 - d_2^2 = 2d_2\sigma\sqrt{T} + \sigma^2 T$$

Substituting the definition of $d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}}$ into the exponent:

$$\begin{aligned} \frac{1}{2}(d_2^2 - d_1^2) &= -d_2\sigma\sqrt{T} - \frac{1}{2}\sigma^2 T = -\left(\ln(S_0/K) + \left(r - \frac{1}{2}\sigma^2\right)T\right) - \frac{1}{2}\sigma^2 T \\ &= -\ln(S_0/K) - rT \end{aligned}$$

Thus, the ratio becomes:

$$\frac{\phi(d_1)}{\phi(d_2)} = \exp(-\ln(S_0/K) - rT) = \frac{K}{S_0} e^{-rT} \Rightarrow S_0 \phi(d_1) = Ke^{-rT} \phi(d_2) \quad (50)$$

This proves that the bracketed expression of equation (49) is zero. The expression for Vega simplifies to just the remaining term:

$$v_{call} = Ke^{-rT} \phi(d_2) \sqrt{T} \quad (51)$$

Using the identity $S_0 \phi(d_1) = Ke^{-rT} \phi(d_2)$ one last time to express the result in terms of S_0 , we obtain the standard formula $v_{call} = S_0 \sqrt{T} \phi(d_1)$. Therefore, the final analytical formula for the absolute payoff is:

$$v^a = 2S_0 \sqrt{T} \phi(d_1) \quad (52)$$

Vega of the Quadratic Payoff

We derive the Vega for the quadratic payoff by differentiating the closed-form pricing equation $V^q = S_0^2 e^{(r+\sigma^2)T} - 2KS_0 + K^2 e^{-rT}$ with respect to σ . Unlike the complex derivation for the straddle, this partial differentiation is straightforward as only the first term depends on σ :

$$\begin{aligned} v^q &= \frac{\partial}{\partial \sigma} (S_0^2 e^{(r+\sigma^2)T} - 2KS_0 + K^2 e^{-rT}) \\ v^q &= S_0^2 \frac{\partial}{\partial \sigma} (e^{(r+\sigma^2)T}) = S_0^2 e^{(r+\sigma^2)T} (2\sigma T) \\ v^q &= 2\sigma T S_0^2 e^{(r+\sigma^2)T} \end{aligned} \quad (53)$$

This result mathematically confirms that the quadratic contract is a much more aggressive bet on volatility than the straddle. While the straddle's Vega scales linearly with the stock price S_0 , the quadratic Vega scales with the square of the stock price (S_0^2) and is directly proportional to the volatility σ itself. This convexity means that as volatility increases, the sensitivity to further volatility increases, creating a compounding effect on the option value.

2.4.3 Graphical Analysis and Interpretation

To visualize the behavior of the option prices, we modeled the sensitivity of the absolute (V^a) and quadratic (V^q) contracts to variations in the market parameters (S_0, K, T, σ). The results are summarized in Figure 4 below.

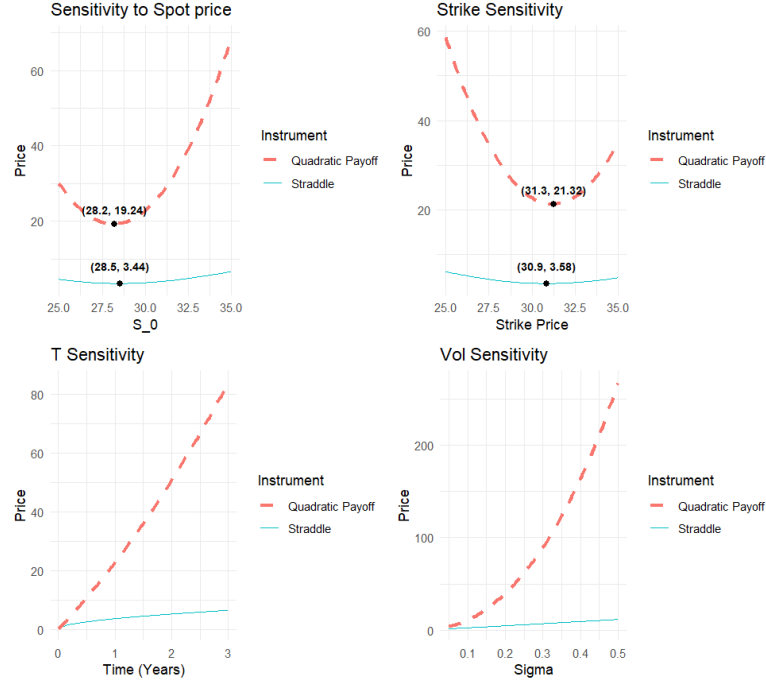


Figure 4: Option Price Sensitivity to Market

A key feature of the sensitivity plots is the symmetry between the asset price and the strike price. The sensitivity to the spot price appears as a mirror image of the strike price sensitivity. This relationship is mathematically expected given the structure of the payoff functions, $|S_T - K|$ and $(S_T - K)^2$, where S_T and K are additive inverses. Consequently, a unit increase in S_0 has an approximately opposite effect on intrinsic value to a unit increase in K .

We can also observe that the minimum value for both option functions does not occur exactly when $S_0 = K$ (ATM), but rather when the spot price is slightly lower than the strike. This shift is caused by the discount factor and the drift. Since the strike price represents a future cash flow, its present value is discounted by the risk-free rate, shifting the equilibrium point. We can mathematically verify this minimum for the quadratic option by solving for the price level where the Delta is zero (the turning point of the convex price curve). Recalling our derived formula for Δ^q :

$$\Delta^q = 2S_t e^{(r+\sigma^2)\tau} - 2K \quad (32)$$

We set $\Delta^q = 0$ and solve for S_t :

$$2S_t e^{(r+\sigma^2)\tau} = 2K$$

$$S_t = K e^{-(r+\sigma^2)\tau}$$

Inputting our project parameters ($K = 30, r = 0.04, \sigma = 0.15, \tau = 1$):

$$S_t = 30 \times e^{-(0.04+0.15^2)(1)} = 30 \times e^{-0.0625} \approx 28.18$$

This analytical result indeed matches our graphical observation perfectly. When $S_t > 28.18$, the price rises because the expected future price S_T is drifting further away from the strike in the upward direction, compounded by the drift terms. On the other hand, when $S_t < 28.18$, the price rises because the current spot is sufficiently low that, even with expected growth, the terminal price is likely to end up below the strike.

The price vs. time graph demonstrates that the value of both instruments increases as the time to maturity (T) increases. This trend is particularly pronounced for the quadratic payoff, reflecting the higher probability of extreme price swings over longer horizons. As explained in the Theta section, the value erodes as time passes, with decay accelerating drastically as $T \rightarrow 0$ and the option loses all extrinsic value.

Finally, the price vs. volatility graph confirms that both instruments are long volatility (positive Vega). However, the quadratic payoff is significantly more sensitive. Since the quadratic contract rewards extreme deviations exponentially rather than linearly, its price curve convexly accelerates as uncertainty (σ) rises, confirming its nature as a more aggressive volatility instrument.

2.5 The Delta (Δ_t) Analytical Computation

The Delta of an option represents the sensitivity of the option price with respect to the underlying asset price. It is defined as the first partial derivative of the pricing function:

$$\Delta_t = \frac{\partial V_t}{\partial S_t} \quad (54)$$

2.5.1 The Absolute Payoff Delta

$$\psi_T^a = |S_T - K| \quad (1)$$

As seen in the section 2.4, we have:

$$\Delta_t^{call} = \frac{\partial C_t}{\partial S_t} = \Phi(d_1) \quad (28)$$

$$\Delta_t^{put} = \frac{\partial P_t}{\partial S_t} = -\Phi(-d_1) = -(1 - \Phi(d_1)) \quad (29)$$

$$\Delta_t^a = \Delta_t^{call} + \Delta_t^{put} = \Phi(d_1) - (1 - \Phi(d_1)) = 2\Phi(d_1) - 1 \quad (30)$$

We know that Delta $\in [0,1]$ for call options and $\in [-1,0]$ for put options. In our case the absolute payoff $|(S_T - K)|$ is a straddle (call + put). Absolute Delta will thus range between $[-1,1]$. This behavior is clearly illustrated in Figure 5 (the right plot), where the

green curve asymptotes toward these limits as the asset price deviates significantly from the strike. Since the standard normal CDF Φ ranges from 0 to 1 (see left plot of Figure 5), the absolute Delta Δ_t^a is strictly bound in $[-1, 1]$.

Behavior of the Absolute Delta

At the start of the contract ($t=0$), with the project parameters set at $S_0 = 30$ and $K = 30$, the option is considered At-The-Money (ATM). As shown in the near Delta-Neutral Zone (the red square) of Figure 5 right plot, the probability of the underlying asset finishing above the strike is slightly higher than the probability of it finishing below it due to the risk-free rate of $r = 4$ meaning $\Phi(d_1) \approx \Phi(0.3417) \approx 0.6337$, rather than the 0.5 value typically associated with a perfectly symmetric distribution of returns around the strike price. Such a symmetry, leading to $\Phi(d_1) = 0.5$, would only occur in a zero-interest-rate environment ($r=0$) with negligible volatility, or more precisely, if the risk-free rate were set to $r = -0.5\sigma^2$ to perfectly offset the convexity adjustment in the BSM drift term.

Consequently, the initial Delta ($\Delta_t^a = 2\Phi(d_1) - 1$) is approximately 0.27, placing the bank in a near-neutral position. This neutrality implies that for very small movements in the asset price around the strike, the increase in the Call's value is almost perfectly offset by the decrease in the Put's value, leaving the overall option price V_0 temporarily stable.

However, as the asset price S_t begins to deviate from the strike, the Delta exhibits a directional convergence. If the price S_t increases far above K , the Call component becomes Deep-In-The-Money while the Put becomes worthless, causing Δ_t^a to approach 1. In this scenario, the option mimics a long stock position, where the option price increases by approximately \$1 for every \$1 increase in the underlying. Conversely, if S_t decreases far below K , the Put dominates and the Delta approaches -1, causing the option to behave like a short stock position. In this case, the option price increases by \$1 for every \$1 decrease in the asset price, fulfilling the client's objective to profit from heavy variations in either direction.

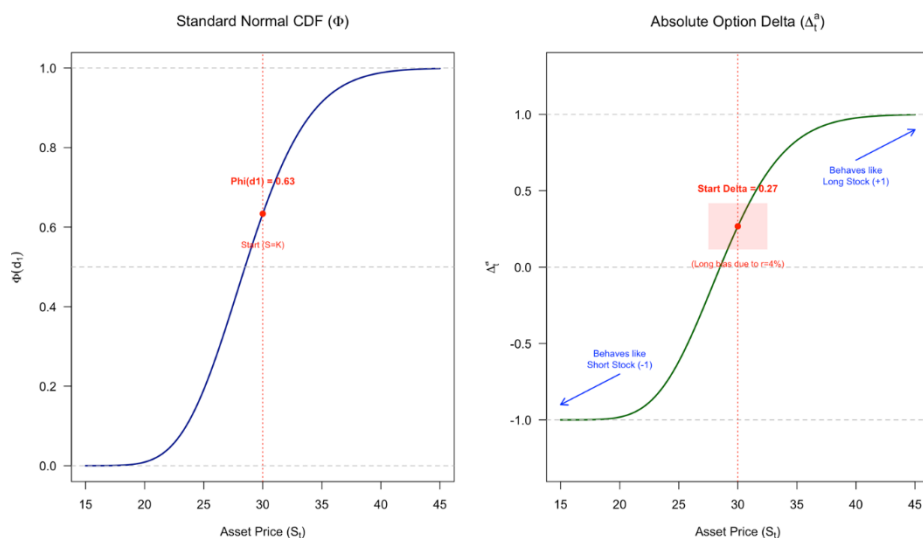


Figure 5: Transformation of the CDF: From $\Phi(d_1)$ to $2\Phi(d_1) - 1$

2.5.1 The Quadratic Payoff Delta

$$\psi_T^q = (S_T - K)^2 \quad (2)$$

As seen in section 2.4, we have:

$$V_t^q = S_t^2 e^{(r+\sigma^2)\tau} - 2KS_t + K^2 e^{-r\tau} \quad (31)$$

$$\frac{\partial V_t^q}{\partial S_t} = \Delta_t^q = 2S_t e^{(r+\sigma^2)\tau} - 2K \quad (32)$$

Behavior of the Quadratic Delta

Unlike the absolute payoff Delta, which is constrained by the standard normal CDF within the range of $[-1, 1]$ (see right plot in Figure 6), the quadratic Delta is a linear function of the underlying asset price S_T and is theoretically unbounded (see left plot in Figure 6). Note that the Delta of the quadratic option is a linear function of S_T , which implies that the Gamma Γ_t^q is constant with respect to the share price. This indicates that the convexity of the option's value remains uniform regardless of whether the asset price moves, providing a consistent second-order sensitivity.

The quadratic payoff, mathematically defined as $\psi_T^q = (S_T - K)^2$, presents a risk profile that is fundamentally distinct from linear vanilla options (a put or a call individually) or the absolute straddle. Derived analytically from the risk-neutral pricing formula, the Delta of this contract Δ_t^q is given by the expression $2S_t e^{(r+\sigma^2)\tau} - 2K$, where $\tau = T - t$ represents the time remaining to maturity. Unlike the absolute payoff Delta, the quadratic Delta is a linear function of the underlying asset price S_t , as clearly illustrated by the straight red line in the left plot of Figure 6, and is theoretically unbounded.

At the inception of the contract ($t=0$), using the project parameters of $S_0 = 30$ and $K = 30$, the option is positioned At-The-Money (ATM). While a standard straddle would be approximately Delta-neutral in this state, the quadratic option exhibits a positive initial Delta. Specifically, using the annualized risk-free rate of 4% and volatility of 15%, the initial Delta Δ_0^q is calculated to be approximately 3.87, a point explicitly marked on the quadratic Delta plot in Figure 6. This non-zero starting point is the result of the positive drift and the convexity adjustment inherent in the squared terminal price representation. For the trader, this means that to hedge a short position in this contract from day one, the bank must immediately purchase approximately 3.87 shares of the underlying asset for every contract sold to the client.

The directional behavior of Δ_t^q as the asset price S_T deviates from the strike reflects the client's desire for an accelerating profit profile. As S_T increases far above K , the Delta continues to rise linearly, depicted as a unbounded linear growth in the left plot of Figure 6, implying that the bank must purchase an increasingly large number of shares to

maintain a hedge as the asset price climbs. Because the Delta is unbounded, the dynamic replication strategy (see section 3.1) is significantly riskier for the bank than that of the absolute option. If S_T explodes, the number of shares required for the hedge becomes massive and it will cost a lot of money for the bank.

Conversely, as S_T decreases significantly below K , the Delta eventually becomes negative. If the asset price S_T approaches zero, the Delta converges toward a theoretical limit of $-2K$, which in this project equals -60 , as shown at the lower intercept in the left plot of Figure 6. This negative Delta indicates that the bank must maintain a short-selling position. A value of -60 implies that for every single contract sold, the bank must be short

60 shares of the underlying asset. This wide and unbounded range of Delta values allows the client to benefit from volatility convexity, where gains accumulate at an accelerating rate as the market exhibits heavy variations.

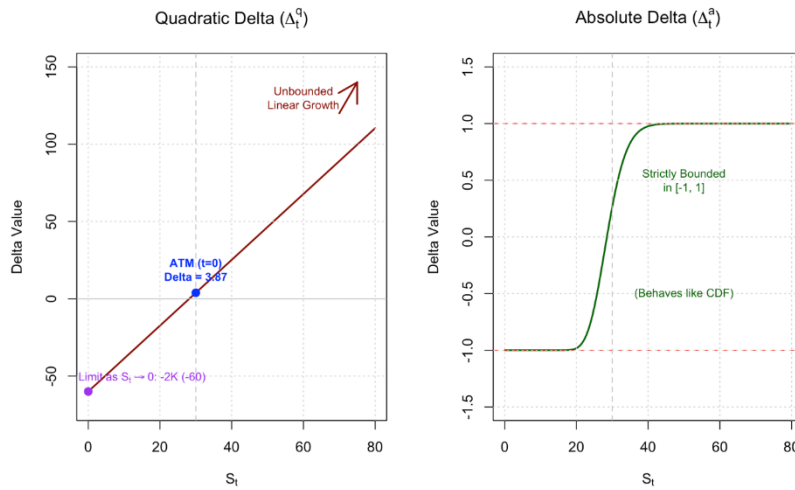


Figure 6: Transformation of the CDF: From $\Phi(d_1)$ to $2\Phi(d_1) - 1$

2.6. Analysis of the Payoff, Option Price, Profit and Delta functions

In order to analyze the dynamics of the options contracts throughout their life cycles, we perform a comparative visualization of their key components. Specifically, we evaluate the terminal payoff ψ_T , the time-dependent price function V_t , the profit function $\psi_T - V_0$ and the option Delta Δ_t . These metrics are illustrated as a function of the underlying asset price S_T .

While the payoff and profit profiles represent static terminal states at T , the price V_t and the delta Δ_t are computed for three distinct time horizons, $t \in (0, T/2, T)$, using a reproducible R implementation to ensure transparency.

2.6.1 Payoff Functions

As displayed in Figure 7, we compare the payoff structures of the quadratic option, the absolute option, and a standard European call.

While both the absolute quadratic payoffs deliver positive returns as S_T moves away from the strike $K=30$ in either direction, their magnitudes differ significantly. The quadratic payoff exhibits strong convexity, meaning gains accumulate at an accelerating rate during heavy price variations, whereas the absolute payoff remains linear.

Comparison with European Call: For $S_T > K$, the European call payoff $\max(0, S_T - K)$ overlaps exactly with the absolute payoff. However, unlike the Straddle and quadratic options, the call provides zero payoff if $S_T < K$, failing to capture gains from downward volatility.

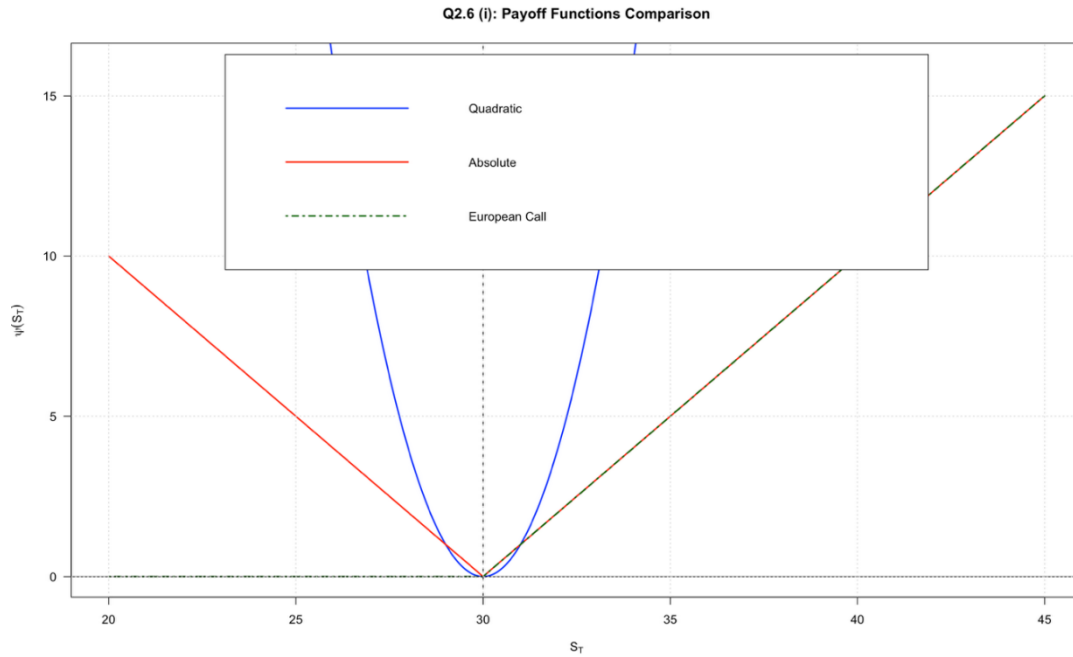


Figure 7: Payoff Functions Comparison

2.6.1 Option Price Functions

As displayed in Figure 8, we analyze the temporal evolution of the price functions V_t for the quadratic option, the absolute Straddle, and the European call across three time horizons: $t \in (0, T/2, T)$.

Initial Valuation and Magnitude: At the inception of the contract ($t=0$), the quadratic option's price is significantly higher ($V_0^q = 22.76$) than those of the absolute Straddle ($V_0^a = 3.64$) and the European call ($V_0^c = 2.41$). This valuation gap is driven by the convexity of the quadratic structure, which heavily weights the probability of extreme terminal asset price deviations in its initial premium.

Temporal Convergence and Time Decay: As the time to maturity $\tau = T - t$ decreases, the price functions (solid and dashed lines) converge toward their respective terminal payoffs (dotted lines). This visual transition illustrates the erosion of the time value

(extrinsic value), by maturity ($t=1$), the option price coincides exactly with its intrinsic payoff, as the uncertainty regarding the terminal spot price S_T is resolved

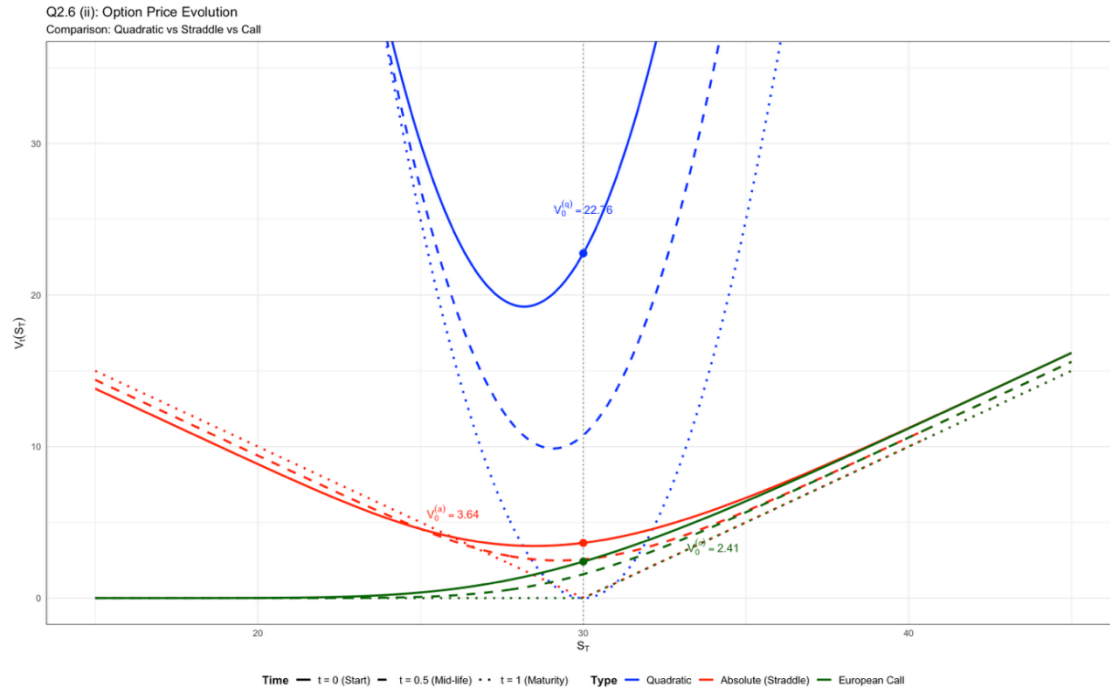


Figure 8: Option Price Evolution

2.6.2 Profit Functions

As displayed in Figure 9, we analyze the profit at maturity, defined as $\psi_T - V_0$ for the quadratic option, the absolute Straddle, and the European call.

Maximum Capital at Risk: For each long position, the maximum loss occurs At-The-Money ($S_T = K = 30$) and corresponds to the total loss of the initial premium V_0 , except for the European Call where the maximum loss is also sustained for any downward variation of S_T below the strike price. In case of $S_T = K$ happening, the quadratic contract represents a significantly larger capital loss ($= -22.76$) compared to the absolute Straddle loss ($= -3.64$) and the European call loss ($= -2.41$), reflecting the higher premium required for its aggressive payoff structure.

The break-even points define the specific asset price variations required to offset the cost of the strategy V_0 . The absolute Straddle breaks even at $S_T = 26.36$ and $S_T = 33.64$. In contrast, the quadratic contract requires much larger heavy variations to recover its substantial initial cost, with break-even hurdles positioned at $S_T = 25.2$ and $S_T = 34.77$. The European call only reaches profitability on the upside, breaking even at $S_T = 32.41$. This is an important point for the client, who must be truly convinced that the asset will skyrocket or collapse (very sharp fluctuations) in this case, exceeding the break-even point in the event of a quadratic contract.

Once the break-even hurdles are breached, the profit of the quadratic contract accelerates due to its inherent convexity. While the straddle profits grow linearly with price moves, the quadratic contract offers a much higher sensitivity to extreme tail

realizations. This visual evidence confirms that the quadratic contract is the optimal choice for a client anticipating massive market shocks, as the profit potential far exceeds linear strategies once the volatility threshold is reached.

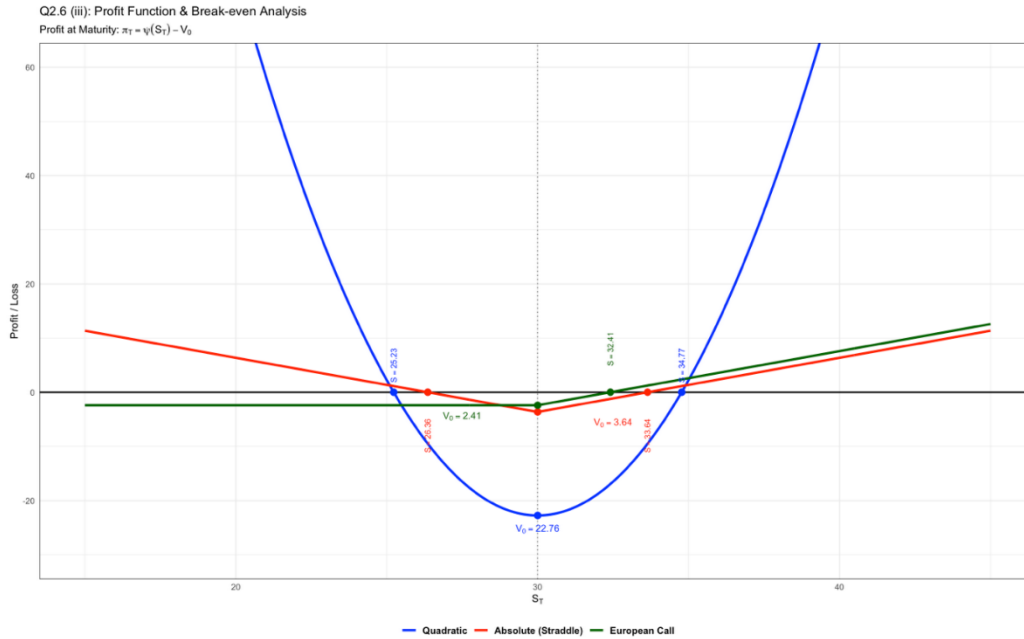


Figure 9: Option Price Evolution

2.6.3 Delta Functions

As shown in Figure 10, we examine the Delta functions ($\Delta_t = \partial V_t / \partial S_t$) across the three time horizons $t \in \{0, 0.5, 1\}$.

The most striking difference lies in the range of the Delta values. While the European call (green curve) and absolute straddle (red curve) are strictly bounded within $[0, 1]$ and $[-1, 1]$ respectively, the quadratic Delta (blue curve) exhibits linear and theoretically unbounded behavior. This confirms that the quadratic contract requires an increasingly aggressive hedging response as the asset deviates from the strike, whereas the Straddle's Delta hedge saturates at ± 1 share.

At the start of the contract, when the asset are At-The-Money ($S_0 = K = 30$), Figure 10 shows that the quadratic Delta starts at a significantly higher level ($\Delta_0^q = 3.87$) compared to the European call ($\Delta_0^c = 0.63$) and the Straddle ($\Delta_0^a = 0.27$). This high initial Delta implies that the bank must immediately purchase nearly four shares for every quadratic contract sold to maintain a Delta-neutral position.

As maturity approaches ($t \rightarrow 1$), the dotted curves in Figure 10 show that the Delta functions become much steeper and more abrupt around the strike price. For the European call, the Delta gradually approaches a Heaviside step function, meaning it moves quickly from 0 to 1 at the strike. Similarly, the straddle's Delta becomes increasingly sharp and eventually behaves like a sign function, jumping almost instantaneously from -1 to $+1$ when the underlying price crosses the strike. For the

quadratic option, the Delta at maturity ($t=1$) becomes the derivative of the payoff = $2(S_T - K)$, which is a straight line with a slope of 2 passing through the strike $K = 30$.

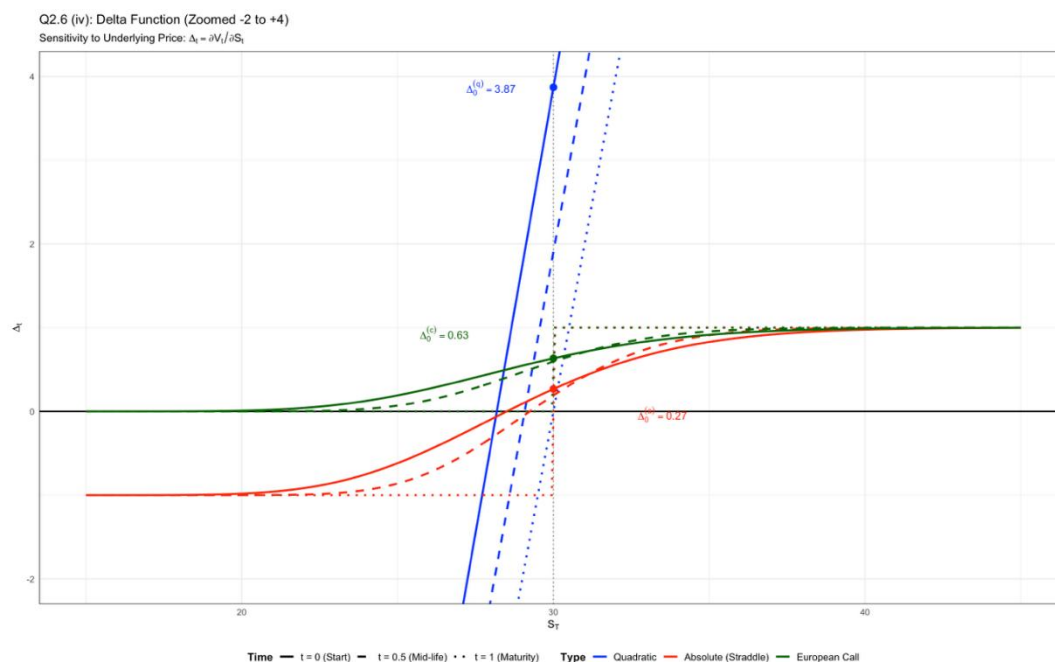


Figure 10: Delta Function (Zoomed -2 to +4)

3. Dynamic Replication of Option Payoffs

3.1 Implementation of a Dynamic Replication Strategy

The objective of section 3.1 is to approximately replicate the payoff of a derivative at maturity using a self-financing dynamic trading strategy involving one risky asset and one risk-free bank account.

We assume $S_0 = K = 30$, maturity $T = 1$ year, volatility $\sigma = 0.15$, and risk-free rate $r = 4$. The theoretical initial prices of both payoffs are obtained analytically. Since trading occurs at discrete times, replication is not exact and leads to a hedging (discretization) error.

We work under the Black–Scholes–Merton framework. Under the risk-neutral measure $\mathbb{P}^*$, the underlying asset price S_t follows a geometric Brownian motion:

$$dS_t = rS_t dt + \sigma S_t dW_t^*, \quad (8)$$

The replication strategy is implemented on an equally spaced time grid:

$$t_i = i\delta, i = 0, \dots, n, \delta = \frac{T}{n}$$

A standard and realistic choice is $n = 252$, corresponding approximately to daily rebalancing over one year each business day. Increasing the rebalancing frequency reduces the discretization error and improves replication accuracy.

3.1.1 Simulation of the underlying asset paths

To evaluate the replication strategy, trajectories of the underlying asset are simulated using the Euler discretization of the GBM:

$$S_{t_{i+1}} = S_{t_i}(1+r\delta+\sigma\sqrt{\delta}Z_i), Z_i \sim \mathcal{N}(0,1). \quad (55)$$

Although an exact discretization of the GBM exists, the Euler scheme is used as specified in the project instructions. The simulated paths are employed exclusively to assess the replication strategy and not for option pricing.

3.1.2 Replication principle: delta hedging

Under the Black–Scholes model, the price of a European payoff at time t can be written as:

$$V_t = v(t, S_t), \quad (56)$$

where $v(t, S)$ is a deterministic pricing function.

The replication strategy relies on delta hedging, where the number of shares held at time t is given by:

$$\Delta_t = \frac{\partial v}{\partial S}(t, S_t) \quad (57)$$

The delta measures the sensitivity of the derivative price to small changes in the underlying asset price.

Even though the option price and the corresponding delta are known exactly through analytical formulas, the replication strategy itself is a dynamic trading procedure that unfolds over time and depends on how the underlying asset price actually evolves. To mimic this market evolution, we simulate possible price paths of the underlying using Monte Carlo methods. Along each simulated path, the hedge ratio is updated at every rebalancing date according to the analytical delta. This allows us to illustrate concretely how the replication strategy operates in practice and to assess the impact of discrete rebalancing on the final hedging error.

3.1.3 Analytical pricing and delta formulas

Absolute payoff

The absolute payoff can be decomposed as:

$$|S_T - K| = (S_T - K)^+ + (K - S_T)^+ \quad (16)$$

defined as the sum of a call and a put with identical strike and maturity.

The pricing function is therefore:

$$v^a(t, S) = C(t, S) + P(t, S) \quad (58)$$

where C and P denote the Black–Scholes prices of the call and the put.

Let

$$d_1 = \frac{\ln(S/K) + \left(r + \frac{1}{2}\sigma^2\right)(T - t)}{\sigma\sqrt{T - t}}$$

The delta of the absolute payoff is:

$$\Delta^a(t, S) = 2N(d_1) - 1 \quad (59)$$

For the absolute payoff, the payoff function is not differentiable at the strike K . This implies that, if $S_T = K$ at maturity, the delta is not uniquely defined. In practice, since no further hedging is possible once the payoff is realized, the replication strategy is stopped at the last rebalancing date before maturity, and the terminal portfolio value is directly compared to the payoff.

Quadratic payoff

The price of the quadratic payoff is given by the risk-neutral valuation formula:

$$v^q(t, S) = e^{-r(T-t)} \mathbb{E}^*[(S_T - K)^2 \mid S_t = S] \quad (60)$$

Under GBM dynamics, the conditional moments are:

$$\mathbb{E}^*[S_T | S_t = S] = S e^{r(T-t)}, \mathbb{E}^*[S_T^2 | S_t = S] = S^2 e^{2r(T-t) + \sigma^2(T-t)}$$

This yields the closed-form pricing function:

$$v^q(t, S) = e^{-r(T-t)} (S^2 e^{2r(T-t) + \sigma^2(T-t)} - 2KS e^{r(T-t)} + K^2) \quad (61)$$

The corresponding delta is:

$$\Delta^q(t, S) = 2S e^{(r+\sigma^2)(T-t)} - 2K$$

Unlike the absolute payoff, the quadratic payoff is smooth and differentiable everywhere, including at maturity.

3.1.4 Portfolio decomposition

At each rebalancing date t_i , the value of the replicating portfolio is decomposed into two components: a position in the underlying asset and a cash account,

$$\Pi_{t_i} = \Delta_{t_i} S_{t_i} + B_{t_i} \quad (62)$$

where Δ_{t_i} denotes the number of shares held in the stock and B_{t_i} the amount invested in the bank account.

This decomposition makes explicit that the replication strategy relies solely on reallocating wealth between the stock and the cash account. At a rebalancing date, the hedge update modifies the portfolio composition but not its total value. In the absence of transaction costs, the strategy is therefore self-financing:

$$\Pi_{t_i^+} = \Pi_{t_i^-} \quad (63)$$

meaning that rebalancing only transfers wealth between the stock position and the cash account.

This property is central to the replication argument: without market frictions, no external cash injection is required, and the portfolio dynamics are entirely driven by the underlying price evolution and the hedge updates.

At maturity, the replication strategy stops at the last rebalancing date t_{n-1} . The terminal portfolio value Π_T , obtained after the final accrual of the cash account and the last stock position, is then directly compared to the contract payoff in order to define the replication error.

3.1.5 Replicating Portfolio Dynamics

This section details how the replicating portfolio evolves over time, distinguishing three key steps: initialization, dynamics between rebalancing dates, and termination at maturity.

Initialization

At the initial date $t_0 = 0$, the replicating portfolio is set up to match the theoretical option price.

The initial stock position is given by the option delta,

$$\Delta_{t_0} = \frac{\partial v}{\partial S}(0, S_0) \quad (64)$$

and the cash account is chosen such that the total portfolio value equals the option price:

$$B_{t_0} = v(0, S_0) - \Delta_{t_0} S_0 \quad (65)$$

As a result, the initial portfolio value satisfies

$$\Pi_{t_0} = \Delta_{t_0} S_0 + B_{t_0} = v(0, S_0) \quad (66)$$

Update between two rebalancing dates

Between two consecutive rebalancing dates t_i and t_{i+1} , the portfolio composition is kept constant.

The stock position Δ_{t_i} remains unchanged, while the cash account accrues at the risk-free rate:

$$B_{t_{i+1}} = B_{t_i} e^{r\Delta t} \quad (67)$$

At the rebalancing date t_{i+1} , the hedge is updated according to the new delta,

$$\Delta_{t_{i+1}} = \frac{\partial v}{\partial S}(t_{i+1}, S_{t_{i+1}}) \quad (68)$$

In the absence of transaction costs, this update is self-financing: the portfolio value does not change during rebalancing, and wealth is only transferred between the stock and the cash account. When transaction costs are introduced, the cash account is reduced by the cost associated with the traded volume, breaking the self-financing property.

Termination at maturity

The replication strategy stops at the last rebalancing date t_{n-1} . After the final accrual of the cash account and the last stock position, the terminal portfolio value Π_T is obtained. This value is then directly compared to the realized payoff $\psi(S_T)$ which gives the hedging error defined as:

$$\varepsilon = \Pi_T - \psi(S_T) \quad (69)$$

In continuous time, the Black–Scholes framework predicts perfect replication. In discrete time, the hedging error is generally non-zero but decreases as the rebalancing frequency increases.

In part “3.3 Comparison of Replication Performance”, this hedging error is used as the main performance measure to conduct a detailed analysis of the replication strategy.

3.1.6 Replicating portfolio dynamics over multiple rebalancing periods

Time	Action	Stock price	Shares held	Equity value	Cash value	Portfolio value
t_0	Initialization	S_0	–	0	$\Pi_0 = v(0, S_0)$	Π_0
t_0	After hedge	S_0	Δ_0	$\Delta_0 S_0$	$B_0 = \Pi_0 - \Delta_0 S_0$	Π_0
t_1^-	Before rebalance	S_1	Δ_0	$\Delta_0 S_1$	$B_0 e^{r\delta}$	$\Pi_1^- = \Delta_0 S_1 + B_0 e^{r\delta}$
t_1	Rebalance	S_1	Δ_1	$\Delta_1 S_1$	$B_1 = \Pi_1^- - \Delta_1 S_1$	$\Pi_1 = \Pi_1^-$
t_2^-	Before rebalance	S_2	Δ_1	$\Delta_1 S_2$	$B_1 e^{r\delta}$	$\Pi_2^- = \Delta_1 S_2 + B_1 e^{r\delta}$
t_2	Rebalance	S_2	Δ_2	$\Delta_2 S_2$	$B_2 = \Pi_2^- - \Delta_2 S_2$	$\Pi_2 = \Pi_2^-$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$t_n = T$	Maturity	S_T	Δ_{n-1}	$\Delta_{n-1} S_T$	$B_{n-1} e^{r\delta}$	Π_T

Figure 11: step-by-step portfolio evolution (refined with ChatGPT-5)

3.2 Impact of Transaction Costs on Replication

In section 3.1, the replication strategies for the absolute payoff and the quadratic payoff were derived under the assumption of a frictionless market. In that setting, the portfolio is self-financing and is rebalanced at discrete dates by adjusting the stock position from $\Delta_{t_{i-1}}$ to Δ_{t_i} , with the remaining wealth invested in the cash account. Transaction costs were neglected, so the replication error was driven only by time discretization.

We now extend this framework by introducing proportional transaction costs. Each time the portfolio is rebalanced, changing the stock holding is no longer free. At rebalancing date t_i , the change in the number of shares is $\Delta_{t_i} - \Delta_{t_{i-1}}$. Given the stock price S_{t_i} , this adjustment corresponds to a dollar amount traded equal to:

$$S_{t_i} \mid \Delta_{t_i} - \Delta_{t_{i-1}} \mid$$

If transaction costs are proportional with rate c (here $c = 50\text{bps}$, i.e. 0.5 per dollar traded), the cost paid at date t_i is

$$TC_i = c S_{t_i} \mid \Delta_{t_i} - \Delta_{t_{i-1}} \mid \quad (70)$$

This cost must be deducted from the cash account at each rebalancing date.

3.2.1 Effect on Portfolio Dynamics and Hedging Error

Relative to the frictionless case studied in section 3.1, the hedging rule for the stock position Δ_t remains unchanged when transaction costs are introduced. However, the portfolio dynamics are significantly affected through the cash account. At each rebalancing date, transaction costs are incurred when adjusting the stock position, leading to a systematic reduction of the cash account over time.

As a consequence, the terminal value of the replicating portfolio is lower than in the frictionless setting, and the distribution of the hedging error is shifted downward. Even if the discretization error were centered around zero in the absence of transaction costs, the presence of market frictions induces a negative bias in the terminal hedging error, since transaction costs are always non-negative and accumulate throughout the trading period.

This effect is illustrated in Figures 12 and 13. Figure 12 shows that, along a given price path, the replicating portfolio with transaction costs consistently lies below both the frictionless replication and the payoff function. Figure 33 further highlights this phenomenon by displaying the gap $\Pi_t - \psi(S_t)$, which becomes increasingly negative over time when transaction costs are included. These graphical illustrations confirm that transaction costs not only increase the magnitude of the hedging error but also introduce a systematic downward drift in the replication performance.

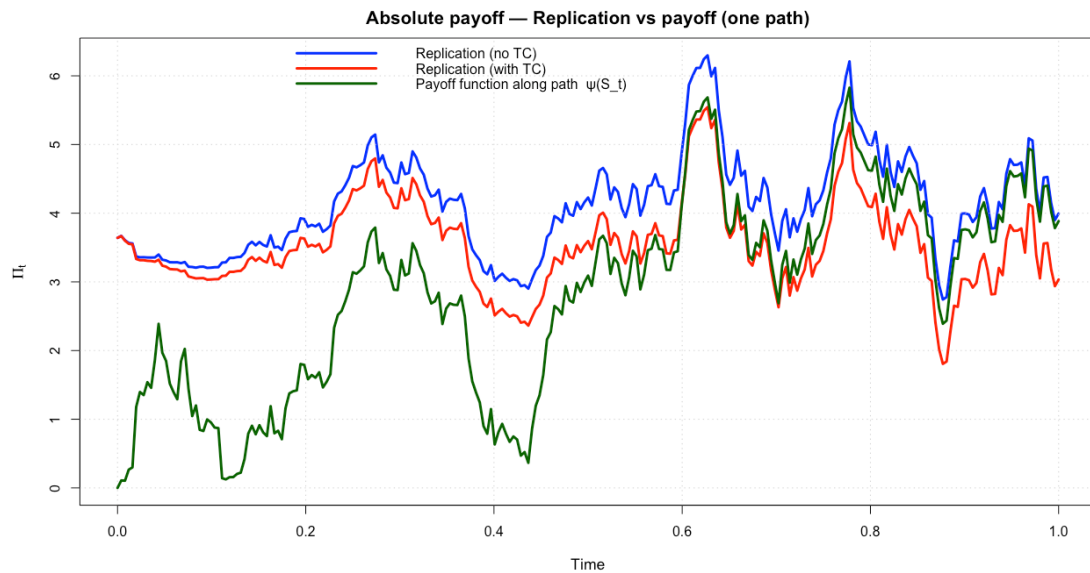


Figure 12: Absolute payoff - Replication vs payoff

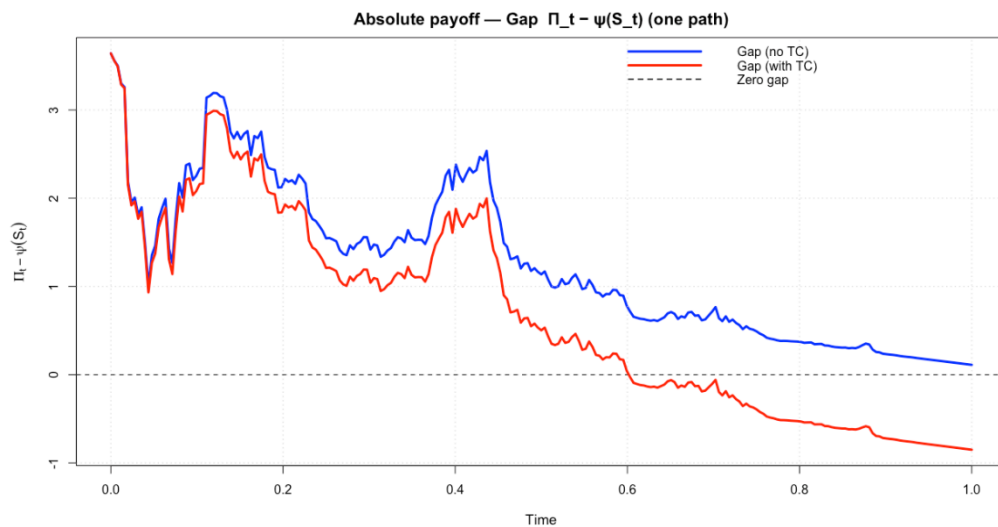


Figure 12: Absolute payoff - Gap (Π_t -Payoff)

3.2.2 Rebalancing Frequency and Cost–Accuracy Trade-off

Transaction costs also fundamentally change the role of the rebalancing frequency n . In section 3.1, increasing n improves replication accuracy by reducing discretization error. When transaction costs are present, increasing the rebalancing frequency creates two opposing effects: it reduces discretization error by tracking the theoretical hedge more closely, but simultaneously increases the cumulative dollar amount traded, thereby raising the total transaction costs. This creates a clear trade-off between accuracy and costs. As a result, rebalancing as frequently as possible is no longer optimal, and an intermediate value of n may minimize the total hedging error.

Finally, while transaction costs affect all replication strategies, their impact depends on the nature of the payoff function. The preceding analysis provides a first intuition, but does not fully capture the mechanisms through which transaction costs interact with the replication strategy. In the next section, we therefore investigate in greater depth the effect of transaction costs on the replication of the absolute and quadratic payoffs. This analysis combines numerical simulations and graphical evidence to compare their sensitivity to transaction costs and to highlight the role played by the shape of the payoff.

3.3 Comparison of Replication Performance

To compare the replication performance of the absolute payoff and the quadratic payoff, we rely on a Monte Carlo simulation. The objective of this code is to evaluate, in a systematic and quantitative way, which payoff can be replicated more accurately in practice when dynamic hedging is performed in discrete time.

The code first simulates a large number of possible price paths for the underlying asset under the risk-neutral measure. More precisely, the underlying price S_t is assumed to follow a geometric Brownian motion and is simulated using an Euler discretization scheme. A total of 20,000 independent paths are generated for the terminal price S_T , ensuring that the results are statistically robust.

For each simulated path, a dynamic replication strategy is then applied, as constructed in section 3.1. The replication is performed by discretely rebalancing a self-financing portfolio composed of the underlying asset and a cash account. The hedging strategy is implemented for several rebalancing frequencies n , namely $n = 52$, 252, and 1000, corresponding to weekly, daily, and very high-frequency rebalancing. This allows us to analyze the impact of the rebalancing frequency on replication accuracy.

The entire procedure is carried out separately for the two payoffs structures. For each payoff and each value of n , the terminal value of the replicating portfolio Π_T is compared to the true payoff $\psi(S_T)$, and the hedging error $\varepsilon_T = \Pi_T - \psi(S_T)$ is computed for every simulated

path. The distribution of hedging errors is then analyzed using several statistical indicators, including the mean error, standard deviation, mean absolute error (MAE), root mean squared error (RMSE) and quantiles.

In addition, the code produces multiple graphical diagnostics, such as density plots, boxplots, and RMSE curves. These visual tools complement the numerical results and provide intuition about the quality and stability of the replication strategies.

Based on these outputs, we can now compare the replication performance of the absolute and quadratic payoffs and determine which payoff is more robust to discretization effects and trading intensity in a dynamic hedging framework.

3.3.1 Interpretation of the replication error statistics

For each payoff structure and rebalancing frequency, several summary statistics of the replication error

$$\varepsilon = \Pi_T^{rep} - \psi_T \quad (71)$$

are computed based on a large number of simulated price paths.

In this section, the replication error is analyzed using several summary statistics. These statistics are computed for each payoff structure and each rebalancing frequency based on a large number of simulated price paths. All numerical values discussed in this section are reported in detail in the summary tables provided in the Appendix, which serve as a quantitative reference for the interpretations presented below.

General effects of rebalancing frequency and transaction costs

Two systematic effects are observed across all error measures. First, in a frictionless market (no transaction cost), increasing the rebalancing frequency reduces discretization error and improves replication accuracy, leading to smaller and more concentrated errors. Then, when transaction costs are introduced, replication errors become systematically more negative and more dispersed, and higher rebalancing frequencies amplify cumulative trading costs.

In the following, these general effects are taken as given, and the analysis focuses on how the different error measures characterize the replication performance of the absolute and quadratic payoffs.

Mean replication error (Mean(ε))

The mean replication error measures the average bias of the replication strategy, that is, whether the strategy tends to over-replicate or under-replicate the payoff on average.

Without transaction costs, both payoff structures exhibit mean errors close to zero, indicating that neither strategy systematically over-replicates nor under-replicates the

payoff. Differences in replication quality therefore mainly arise from dispersion and tail behavior rather than from average bias.

When transaction costs are introduced, the mean replication error becomes significantly negative for both payoff structures. This negative bias reflects the fact that transaction costs systematically reduce the value of the replicating portfolio at each rebalancing date.

Furthermore, the magnitude of the negative bias increases with the rebalancing frequency n . Although more frequent rebalancing improves replication accuracy in a frictionless market, it leads to higher cumulative trading costs, which dominate the discretization gains and result in a more negative average replication error.

Comparing the two payoff structures, the quadratic payoff exhibits a substantially more negative mean error than the absolute payoff for all values of n . This indicates that the replication strategy associated with the quadratic payoff is more sensitive to transaction costs, due to larger and more frequent adjustments in the hedge ratio.

Standard deviation of the replication error ($SD(\varepsilon)$)

The standard deviation of the replication error measures how much the replication performance varies across different simulated price paths. It captures the stability of the hedging strategy and its sensitivity to the specific trajectory followed by the underlying asset.

For all rebalancing frequencies, the quadratic payoff exhibits a substantially higher standard deviation than the absolute payoff. This indicates that its replication error is more path-dependent and less stable. This higher dispersion is explained by the stronger convexity of the quadratic payoff, which induces larger variations in the hedge ratio and amplifies path-dependent effects.

When transaction costs are introduced, the standard deviation of the replication error increases significantly for both payoff structures. Transaction costs add an additional source of variability, since the cumulative trading costs depend on the magnitude and frequency of hedge adjustments along each path.

Moreover, the dispersion tends to increase with the rebalancing frequency. While frequent rebalancing improves stability in a frictionless setting, it leads to greater variability once transaction costs are accounted for, due to the accumulation of path-dependent trading costs.

Overall, the quadratic payoff remains remarkably less stable than the absolute payoff, highlighting its higher sensitivity to both discretization effects and market frictions.

MAE - Mean Absolute Error

The Mean Absolute Error (MAE) measures the average magnitude of the replication error, independently of its sign. It can be interpreted as the typical replication error observed along a single price path. Although the MAE is interpreted as the typical replication error on a single path, it is computed as an average over a large number of simulated price paths. Each reported MAE therefore summarizes the expected magnitude of the replication error for a randomly realized trajectory of the underlying asset.

For a given rebalancing frequency, the quadratic payoff exhibits a substantially larger MAE than the absolute payoff. This indicates that, on a typical path, the replication error is larger for the quadratic payoff. This behavior is consistent with the higher convexity of the quadratic payoff, which induces stronger sensitivity to underlying price movements and larger hedge adjustments.

When transaction costs are introduced, the MAE increases significantly for both payoff structures. Transaction costs add a systematic loss component to each rebalancing step, increasing the typical deviation between the replicating portfolio and the payoff.

Moreover, the MAE increases with the rebalancing frequency n , since more frequent trading leads to higher cumulative costs. This highlights the trade-off between hedging accuracy and transaction costs in practical replication strategies.

RMSE - Root Mean Squared Error

The Root Mean Squared Error (RMSE) penalizes large replication errors more strongly than the MAE, as it squares the error before averaging. It therefore captures both overall accuracy and exposure to extreme replication outcomes.

For all values of n , the RMSE associated with the quadratic payoff is significantly larger than that of the absolute payoff. This indicates that the quadratic payoff is more exposed to large replication errors in extreme scenarios. The larger gap between RMSE and MAE observed for the quadratic payoff further suggests the presence of heavier tails in the error distribution, reflecting higher sensitivity to extreme price movements of the underlying asset.

When transaction costs are taken into account, the RMSE increases sharply for both payoff structures and grows with the rebalancing frequency. Frequent rebalancing amplifies the accumulation of transaction costs, which not only shifts the error distribution downward but also increases the likelihood of large negative errors in adverse scenarios.

Overall, the RMSE highlights that transaction costs disproportionately affect strategies that require large and frequent hedge adjustments, with the quadratic payoff being particularly sensitive.

3.3.2 Interpretation of replication error quantiles

Quantiles provide a detailed description of the distribution of replication errors by focusing on extreme and representative outcomes. Unlike average measures, they allow us to assess tail risk and to understand how replication performance behaves in adverse and favorable scenarios.

Lower quantiles: downside risk (1% and 5%)

Lower quantiles capture the most adverse replication outcomes and are particularly relevant from a risk-management perspective.

The 1% quantile represents the worst replication losses observed across simulated paths, indicating that only 1% of trajectories lead to errors more severe than this value. In a frictionless setting, increasing the rebalancing frequency improves these lower quantiles for both payoff structures, as extreme under-replication events become less severe when discretization error is reduced. This behavior reflects the convergence toward continuous-time replication.

However, for all rebalancing frequencies, the quadratic payoff exhibits substantially more negative lower quantiles than the absolute payoff. This indicates that, in extreme scenarios, the quadratic payoff is exposed to much more severe under-replication. This difference is explained by the stronger convexity of the quadratic payoff, which amplifies hedging errors during large movements of the underlying asset.

The 5% quantile follows the same qualitative pattern as the 1% quantile but provides a more robust measure of downside risk, as it is less sensitive to very rare paths. Across all configurations, the quadratic payoff consistently displays more negative 5% quantiles than the absolute payoff, confirming that its replication strategy is less robust in adverse scenarios.

When transaction costs are introduced, lower quantiles deteriorate sharply for both payoff structures. The worst-case replication losses become substantially larger, especially for higher rebalancing frequencies. This effect is particularly pronounced for the quadratic payoff, whose extreme losses increase dramatically due to the accumulation of large, path-dependent transaction costs. Overall, lower quantiles clearly show that transaction costs strongly amplify tail risk, especially for strategies requiring frequent and sizable hedge adjustments.

Central tendency: median (50%)

The median represents the central outcome of the replication error distribution, indicating that half of the simulated paths yield a replication error below this value and half above.

In the absence of transaction costs, the median replication error is close to zero for both payoff structures and for all rebalancing frequencies. This confirms that, on a typical path,

the replication strategy neither systematically over-replicates nor under-replicates the payoff.

When transaction costs are introduced, the median becomes strictly negative, indicating a downward shift of the entire error distribution. The magnitude of this shift increases with the rebalancing frequency and is substantially larger for the quadratic payoff. This reflects the cumulative impact of transaction costs on strategies that require frequent hedge adjustments.

Upper quantiles: favorable outcomes (95% and 99%)

Upper quantiles describe favorable replication outcomes, corresponding to scenarios in which the replicating portfolio performs relatively well.

The 95% quantile indicates that only 5% of simulated paths yield replication errors larger than this value, while the 99% quantile focuses on the very best 1% of outcomes. Although these cases may involve over-replication, they are generally less relevant from a risk-management perspective, particularly for an option seller.

In the presence of transaction costs, upper quantiles are typically reduced, indicating that even the most favorable scenarios are negatively affected by cumulative trading costs. Comparing upper and lower quantiles highlights a clear asymmetry in the error distribution: the lower tail is significantly heavier than the upper tail, especially for the quadratic payoff and when transaction costs are present.

This asymmetry confirms that the main risk associated with replication strategies lies in severe under-replication rather than occasional over-replication.

3.3.3 Graphical analysis

Density plots of the replication error ($n = 252$)

For a continuous variable such as the replication error ε , a probability density function (PDF) describes where values tend to concentrate and how likely different ranges of values are. In the present setting, the exact (theoretical) distribution of ε is unknown, because the replication error results from simulated price paths, discrete rebalancing, payoff non-linearity, and (when applicable) transaction costs. For this reason, the density shown in the graphs is an empirical density, i.e. a data-based estimate of the PDF constructed from the simulated replication errors rather than from a closed-form theoretical model.

The two panels display the empirical density of the terminal replication error $\varepsilon = \Pi_T - \psi(S_T)$, for the absolute payoff (solid line) and the quadratic payoff (dashed line). The vertical reference line at $\varepsilon = 0$ corresponds to perfect replication. The location of the density relative to zero indicates a possible bias (over- or under-replication), while the spread and shape of the density reflect the variability of the hedging performance and the risk of large replication errors.

Without transaction costs (noTC): Both empirical densities are centered close to zero, indicating no significant average bias in a frictionless market. However, the density associated with the absolute payoff is highly concentrated around $\varepsilon = 0$, which points to a stable replication strategy with typically small errors. In contrast, the quadratic payoff exhibits a much more dispersed density, with wider tails, showing greater sensitivity to price path variations and a higher likelihood of large replication errors.

With transaction costs (TC): When transaction costs are introduced, the empirical densities shift clearly to the left, meaning that replication errors become systematically negative. This reflects under-replication, as repeated trading costs reduce the terminal value of the hedging portfolio. The effect remains moderate for the absolute payoff, whose density stays relatively concentrated, though slightly below zero. For the quadratic payoff, the shift is much stronger and the density remains highly dispersed, indicating that transaction costs severely deteriorate hedging performance and lead to frequent and substantial shortfalls.

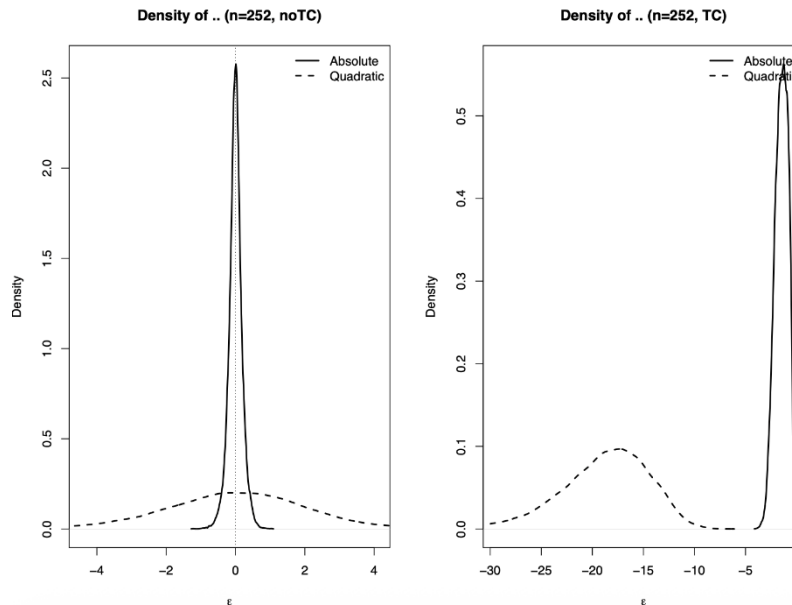


Figure 14: Density of the replication error with and without TC

Boxplots of the replication error

The boxplots summarize the distribution of the terminal replication error $\varepsilon = \Pi_T - \psi(S_T)$ for $n = 252$, with the dashed horizontal line at $\varepsilon = 0$ representing perfect replication. The median indicates the typical bias (above/below zero), the box (IQR) measures dispersion, and the whiskers and outliers reflect tail risk.

No transaction costs (noTC): The absolute payoff exhibits a very tight box centered close to zero, with short whiskers, meaning that replication errors are typically small and stable across paths. In contrast, the quadratic payoff shows a much wider IQR and long whiskers (large spread), indicating substantially higher variability and more frequent large deviations from zero—even though the central tendency remains around $\varepsilon = 0$.

With transaction costs (TC): Both payoffs shift downward: the medians become negative, consistent with a systematic shortfall caused by cumulative trading costs. The absolute payoff remains relatively concentrated (small IQR), suggesting that errors are still controlled but biased slightly below zero. The quadratic payoff is strongly affected: the entire box is far below zero and very wide, with long whiskers, showing both a large negative bias and high dispersion. Overall, transaction costs amplify the under-hedging effect, and the quadratic payoff becomes markedly more expensive to hedge and riskier in terms of replication error.

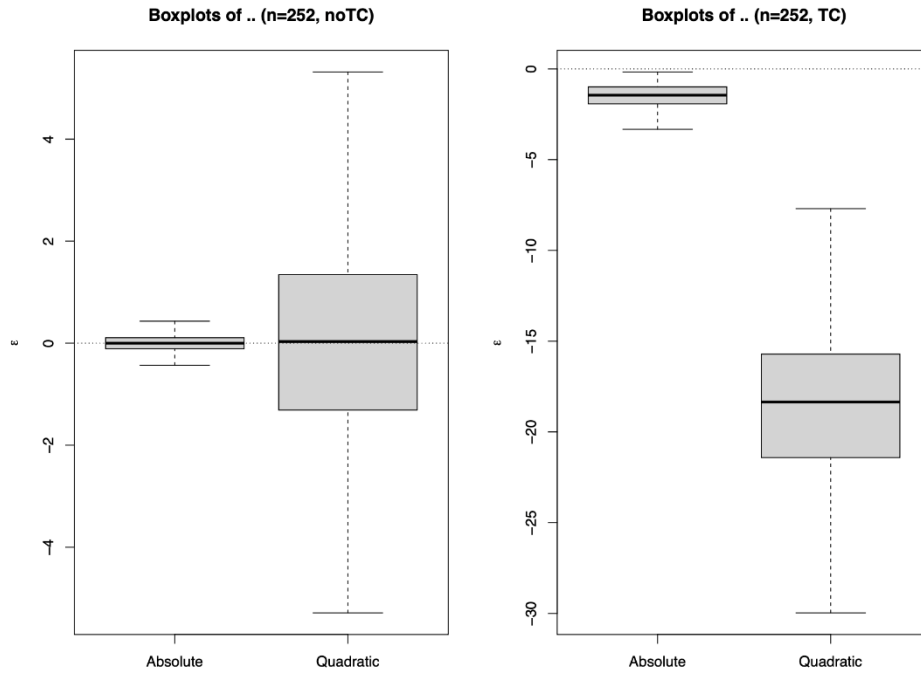


Figure 13: Boxplots of the replication error with and without TC

RMSE vs number of rebalancing intervals

Figure 15 shows the Root Mean Squared Error (RMSE) of the replication error as a function of the number of rebalancing intervals n , for the absolute and quadratic payoffs, both with and without transaction costs.

In the absence of transaction costs, the RMSE decreases as n increases for both payoffs. This confirms that more frequent rebalancing improves hedging accuracy by reducing discretization error. For all values of n , the quadratic payoff exhibits a higher RMSE than the absolute payoff, reflecting its greater sensitivity to price movements and the higher curvature of the payoff function.

When transaction costs are introduced, the behavior changes markedly. For the absolute payoff, the RMSE remains relatively low and only moderately increases with n . In contrast, the quadratic payoff displays a strong increase in RMSE as the rebalancing frequency rises. This indicates that, for highly nonlinear payoffs, the cumulative effect of transaction costs dominates the reduction in discretization error, leading to a deterioration of replication performance when hedging too frequently.

Overall, this figure highlights a clear trade-off between hedging accuracy and transaction costs: while increasing n is beneficial in frictionless markets, it can be counterproductive in the presence of transaction costs, especially for payoffs with high curvature.

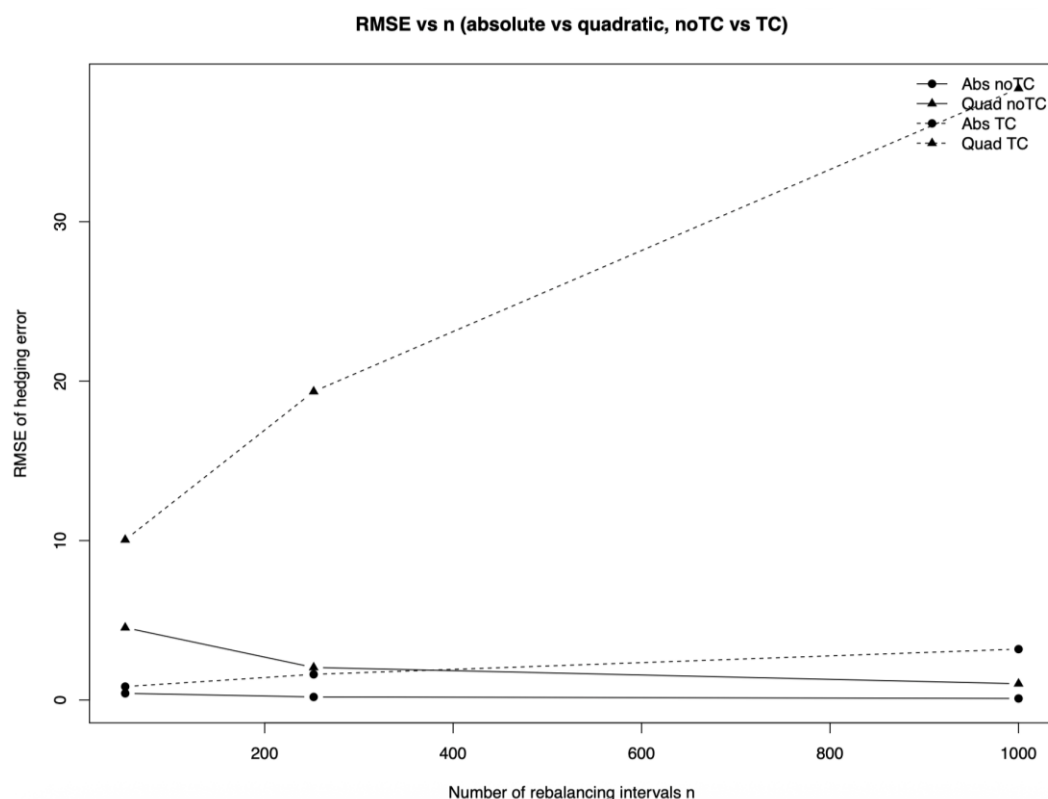


Figure 14: RMSE vs number of rebalancing intervals

Mean transaction cost vs number of rebalancing intervals

Figure 16 presents the mean total transaction cost as a function of the number of rebalancing intervals n for both payoffs, assuming a transaction cost of 50 basis points per dollar traded.

For both payoffs, the mean transaction cost increases monotonically with n , which is expected since more frequent rebalancing leads to higher trading volume. However, the magnitude of this increase differs significantly between the two payoffs. The quadratic payoff generates substantially higher transaction costs than the absolute payoff for all values of n . This reflects the larger and more frequent adjustments of the hedging position required to replicate a payoff with strong curvature.

Combined with the RMSE results, this figure explains why the replication error for the quadratic payoff increases sharply with n in the presence of transaction costs. The rapid growth of transaction costs offsets and eventually outweighs the benefits of finer rebalancing, resulting in poor overall hedging performance.

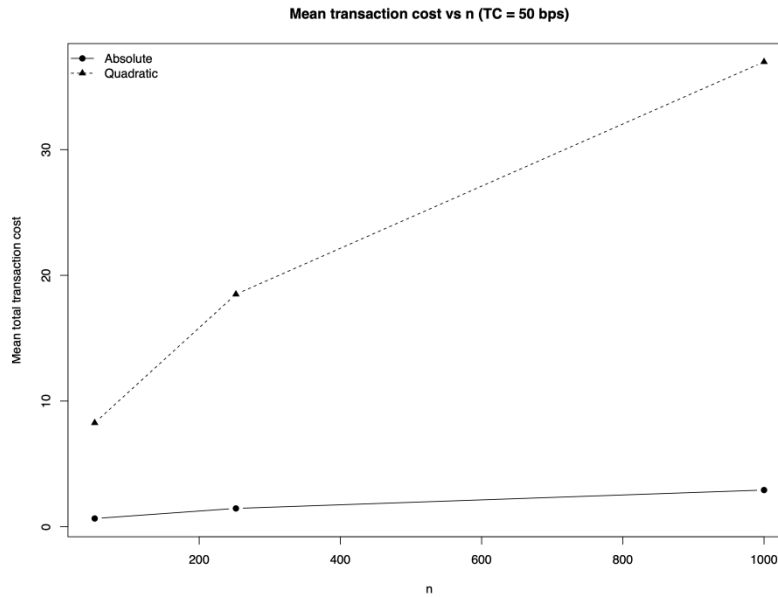


Figure 15: Mean transaction cost vs number of rebalancing intervals

3.4 Chapter 3 Conclusion

This section compared the replication performance of the absolute and quadratic payoff structures using both graphical analysis and numerical error measures across different rebalancing frequencies.

In the absence of transaction costs, increasing the rebalancing frequency systematically improves replication accuracy for both payoffs. Hedging errors become more concentrated around zero and their dispersion decreases, reflecting the reduction of discretization error and the convergence toward the continuous-time replication framework.

Across all rebalancing frequencies, the quadratic payoff exhibits larger dispersion and more pronounced tail behavior than the absolute payoff. This difference is explained by the higher convexity of the quadratic payoff, which induces larger and more frequent adjustments of the hedge ratio, making the replication strategy more sensitive to discretization effects.

When transaction costs are introduced, replication errors become systematically more negative and more dispersed as the rebalancing frequency increases. Although frequent rebalancing reduces discretization error, the cumulative effect of transaction costs dominates and deteriorates replication performance. This effect is significantly stronger for the quadratic payoff due to its higher trading intensity.

Overall, while both payoff structures are theoretically replicable under the Black–Scholes framework, the absolute payoff leads to a more stable and cost-efficient replication strategy in practice. Taking into account discretization effects and transaction costs, the absolute payoff therefore represents the preferable contract to offer to a client.

4. Final Assessment and Strategic Recommendation

In this report, we have conducted a comprehensive evaluation of two derivative products with different payoffs designed to capitalize on the magnitude of asset price movements. Our investigation first addressed the construction and intuition of these trades, demonstrating that while the absolute payoff is a linear combination of two vanilla options, the quadratic payoff requires a more sophisticated static replication strategy. By applying the Carr-Madan spanning formula, we proved that the smooth parabolic curve of a quadratic payoff can be synthesized through a weighted portfolio of Out-of-the-Money options.

We further established a rigorous pricing framework, cross-validating the fair value of these contracts under the Black-Scholes framework using Monte Carlo simulations, numerical integration, and analytical closed-form formulas. These methods consistently confirmed that the quadratic contract commands a significantly higher premium compared to the straddle (22.76 vs. 3.64). This "convexity premium" is driven by the fact that the quadratic payoff assigns exponentially higher weights to large tail deviations, making the contract much more expensive to enter under normal market conditions.

Finally, we assessed operational risks by implementing a dynamic replication of the payoffs using a self-financing strategy involving the underlying asset and a risk-free bank account. Through dynamic Delta-hedging simulations, we evaluated the practical feasibility of maintaining these positions depending on the rebalancing frequency. We found that the quadratic payoff introduces significantly higher hedging risks for the bank, as its Delta is theoretically unbounded and necessitates much more aggressive trading than the absolute payoff. Furthermore, our analysis of transaction costs and discrete rebalancing revealed a systematic downgrade in replication performance for both products. This error is particularly severe for the quadratic strategy, where the high trading intensity required to maintain the hedge can outweigh the benefits of frequent rebalancing.

Final Product Recommendation

Based on our results, the optimal choice for the client depends on her specific convictions regarding future volatility. If the client anticipates significant but realistic volatility, the Absolute Payoff is the more efficient tool. To reach the break-even point for the absolute payoff, the underlying asset must undergo a movement of at least 12.13% (reaching 26.36 or 33.64). While this is already an extreme event, the quadratic payoff requires a 15.9% move (reaching 25.2 or 34.77) just to offset the substantial initial premium.

If these extreme fluctuations do not materialize and the asset remains near the strike price, the client faces a significantly higher capital loss with the quadratic contract, as the initial premium lost could be nearly six times larger than that of the straddle. Therefore,

unless the client is truly convinced of an unprecedented "black swan" event that will trigger explosive squared returns, we recommend the Absolute Payoff. It offers a more attainable path to profitability, and it is a more cost-efficient replication strategy for the bank.

(Note : AI tools were used for coding and text upgrading, as well as for research purposes and explanation of mathematical formulas.)

Bibliography

Brenner, M., & Galai, D. (1989). New Financial Instruments for Hedging Changes in Volatility. *Financial Analysts Journal*. <https://doi.org/10.2469/faj.v45.n4.61>

Brenner, M., & Galai, D. (1993). Hedging Volatility in Foreign Currencies. *The Journal of Derivatives*. <https://doi.org/10.3905/jod.1993.407870>

Carr, P., & Madan, D. (2001). Towards a Theory of Volatility Trading. Cambridge University Press. <https://doi.org/10.1017/CBO9780511569708.013>

Gratton, P. (2025). *Option Greeks: The 4 Factors to Measure Risk*. Investopedia. <https://www.investopedia.com/trading/getting-to-know-the-greeks/>

Vrins, F. D. (2025). *Derivatives Pricing*. Cambridge University Press. <https://doi.org/10.1017/9781009554664>